

ASSIGNMENTS IN **CHEMISTRY** BOOK 3

Peter Gribben & Wendy St Germain



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CHEMISTRY
BOOK 3

Peter Gribben & Wendy St Germain



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INTRODUCTION

This series of chemistry assignments has been written by a highly-experienced team of authors. Each worksheet has been specially designed to meet the needs of busy teachers of chemistry – they require minimal preparation and do not compromise on quality of content.

Most of the material requires very little prior knowledge from the student. New chemical terms are explained as they are met in the text and important concepts are revised or summarised. Areas that students have traditionally found difficult have been broken down into sections and are examined one step at a time.

There are many worked examples of calculations and sufficient questions for students to practise and master difficult concepts. Research questions have been included where appropriate to give students the opportunity to explore the history of chemistry.

CONTENTS

1	POLYMERS 1	4
2	POLYMERS 2	6
3	PROBLEMS WITH PLASTICS?	8
4	THE CARBON CYCLE	9
5	CARBOHYDRATES	10
6	AMINO ACIDS AND PROTEINS	11
7	FERMENTATION	13
8	ETHANOL AND ETHENE	14
9	ESTERS	15
10	THE ATMOSPHERE 1	16
11	THE ATMOSPHERE 2	18
12	ROLES OF CHEMISTS	20
13	SULFUR	21
14	IRON OCCURRENCE AND MANUFACTURE	23
15	THE SOLVAY PROCESS	24
16	REDOX 1	25
17	REDOX 2	27
18	REDOX 3	29
19	REDOX 4	31
20	ACIDS AND BASES 1	33
21	ACIDS AND BASES 2	35
22	ACIDS AND BASES 3	37
23	NUCLEAR CHEMISTRY 1	39
24	NUCLEAR CHEMISTRY 2	41
25	PETROCHEMICALS: REVIEW	43
26	CORROSION OF IRON	44
	ANSWERS	45

The word **polymer** comes from the Greek, meaning **having many parts**. Polymers are large molecules made up of many repeating subunits called **monomers** (*poly*: many; *mono*: one). Consider a metal chain. Imagine a polymer as representing the entire chain and each link as representing one monomer.

Definitions of a polymer

- A molecule consisting of repeated subunits
 - A large molecule formed by the combination of many similar, smaller molecules
 - A molecule of high relative molecular mass, made up of repeating molecules of low relative molecular mass.
- 1 Give another definition of a *polymer*.

Natural polymers

When people refer to polymers, they often mean artificial polymers (also referred to as 'plastics'). There are many natural polymers; common ones include starch, cellulose, proteins, wool, silk, cotton and rubber.

Starch is made up of thousands of repeating glucose units. Glucose is a sugar (saccharide) and its structure is shown in figure 1. Starch is made up of glucose units (slightly altered) and has the formula $(C_6H_{10}O_5)_n$.

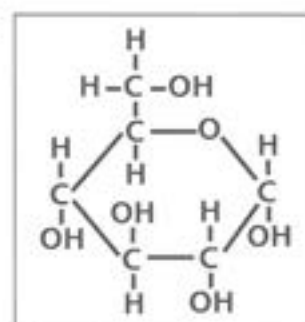


Figure 1

- 2 Copy figure 1. Beside your diagram write the molecular formula of glucose.
- 3 Starch is a polysaccharide. What does *polysaccharide* mean?
- 4 In the formula for starch, $(C_6H_{10}O_5)_n$, what does the *n* stand for?

Many plants store starch as a food source and this energy source can be utilised by humans. Cellulose is another polymer found in plants. It is a long chain polymer with glucose molecules as the monomer, but cellulose is not digestible by humans. The glucose monomers in this polymer are joined in a different way to those in starch. Cellulose has the formula $(C_6H_{10}O_5)_n$ and it is the common material of plant cell walls. It provides the structural strength for natural fibres; e.g. cotton is 95% cellulose.

- 5 Name three plants rich in starch.
- 6 Give two similarities and two differences between starch and cellulose.

Proteins are polymers made up of amino acid monomers. Silk is a natural protein fibre obtained from the cocoons produced by silkworms. Traditionally, plant (and animal fibres) have been spun and woven into cloth.

- 7 Give one use for silk material.
- 8 Name one animal fibre used in clothing.

Natural rubber is obtained from **latex**, a milky emulsion that occurs in the sap of certain tropical plants. The major source of natural rubber is the para rubber tree. When an incision is made into the bark of the tree, latex oozes out. The tree responds to the wound by producing more latex. The latex is treated to form solid rubber and then processed.

- 9 Give two uses for rubber.

Natural rubber has drawbacks, as it is especially sensitive to temperature. It becomes soft when warm and brittle when cold. It also perishes (loses structure and strength) over time. The addition of sulfur to rubber (vulcanisation) was patented by Charles Goodyear in the middle of the nineteenth century. This process stabilises and improves the properties of rubber by cross-linking the individual polymer molecules. Isoprene (2-methyl-1, 3-butadiene) is the major monomer in natural rubber. Its structural formula is shown in figure 2.

- 10 Give the molecular formula of isoprene.
- 11 Give three disadvantages of natural rubber.
- 12 What is *vulcanisation*?
- 13 Which monomer is present in both starch and cellulose?

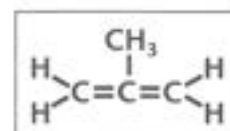


Figure 2

Synthetic polymers

Manufacturers have always been interested in finding alternative materials to limited resources or substances that are expensive to produce. One example is ivory – the tusks (enlarged teeth) of elephants. Ivory is a hard, white, opaque substance.

14 What is meant by *opaque*?

Ivory has been used for billiard balls, ornaments and (white) piano keys. It is a finite resource and many people have concerns about killing animals in order to obtain a decorative substance. In the mid nineteenth century, Alexander Parkes was one of the first people to develop an alternative to ivory. He invented **pyroxylin** which is also known as **Parkesine**. In Parkes's process, cellulose is dissolved and then treated with nitric acid.

15 Name one source of cellulose.

James Wesley Hyatt used cellulose to make **celluloid**. Remember, '-oid' means 'like', therefore celluloid is cellulose-like. The term celluloid refers to a class of compounds made from cellulose, camphor (derived from laurel trees) and nitric acid. One of the first uses of celluloid was making false teeth (dentures). However, the celluloid tasted of camphor and the dentures tended to soften when the wearer drank hot liquids. Celluloid became a replacement for ivory and tortoiseshell. At the beginning of the twentieth century, celluloid was adapted and produced in strip form for the newly-emerging motion picture industry. However, celluloid was very flammable and also tended to decompose over time.

16 Give two uses of ivory.

17 Give two reasons for finding a substitute for ivory.

18 Give two desirable properties of a material used for photographic film.

Cellulose has also been used as the raw material for making fibres for clothing, especially in making artificial 'silks'. Silk has long been a desirable material but it is expensive. The first artificial silks had many of the properties of the real material. However, like celluloid, they were highly flammable. Eventually a safer form of artificial silk was made: rayon. Although rayon has now been superseded by better artificial fibres, it is still in production because it is cheap and can be blended with both natural fibres and other synthetic fibres.

Research 19 What is the source of silk? How is silk thread obtained from this source?

20 Suggest a reason why fibres are blended.

Bakelite™: This artificial polymer is considered the first true 'plastic' because it is not based on any molecules already present in nature. In physics, the word 'plastic' has a different meaning to what you might expect. It is an adjective (or 'describing' word). Using the physics definition below, plastic materials would include plasticine™ and mild steel. When a force is applied to them, these materials change shape and keep the new shape.

21 Give two materials that are 'plastic' according to the physics definition.

22 Give your own definition of *plastic*.

Plastic: the property of a material that causes it to change shape when in contact with a force, and to remain that shape until another force is applied.

Leo Hendrik Baekeland invented Bakelite™ in 1909. He discovered that when phenol and methanal (formaldehyde) were mixed together and heated, a hard solid formed. This solid is a tough, heat-resistant and cheap material that was first used in electrical and mechanical parts. Similar polymers (phenolic resins) are still used to make electronic printed circuit boards.

23 Figure 3 shows the structures of phenol and methanal. Copy the diagrams and write the molecular formula of each substance.

24 Give two properties of bakelite.

In everyday use, we tend to use the term **plastic** to refer to synthetic polymers that can be moulded.

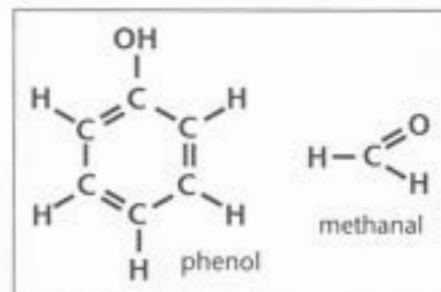


Figure 3

Plastics and heat

Plastics are divided into two major groups according to their behaviour when exposed to heat. **Thermoplastic plastics** soften when heated and harden on cooling. This means that they can be melted down and reformed many times. **Thermosetting plastics** can be moulded when they are initially liquid but once set, do not soften when heated. Bakelite was the first thermosetting plastic.

- 1 Give an example of when a *thermosetting plastic* is more useful than a *thermoplastic plastic*.
- 2 Give an example of when a *thermoplastic plastic* is more useful than a *thermosetting plastic*.

Polyethylene There are two forms of polyethylene (polythene): low-density polyethylene (LDPE) and high-density polyethylene (HDPE). Both polymers have ethylene (ethene) as the monomer. Figure 1 shows the structures of ethylene and polyethylene. The polyethylene molecule consists of thousands of repeating units; the diagram only shows a small part of the chain.

- 3 Give the molecular formula of ethylene.
- 4 The molecular formula of polyethylene is often written as $-(C_2H_4)_n-$. Explain what is meant by this formula.

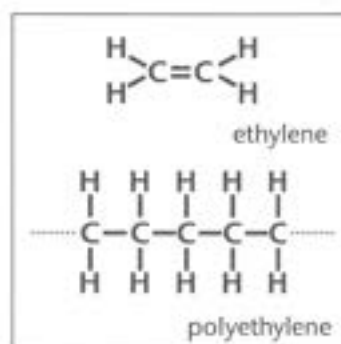


Figure 1

Polyethylene is an **addition polymer**. It is formed by many ethylene molecules joining together when the carbon-carbon double bond breaks, and forms bonds with adjacent molecules. Polyethylene is very cheap, strong, flexible and chemically unreactive. LDPE is used to make packaging materials (especially supermarket shopping bags) and HDPE is used for containers and automotive fittings. Because polyethylene is unreactive, discarded items made of this material don't break down but remain in the environment. If they are collected, they can be recycled. Polythene shopping bags have been handed out by retailers for many years. They are ideal for this purpose, but many people want to reduce their use.

- 5 Define *polymer*.
- 6 Polyethylene is a thermoplastic plastic. What is meant by *thermoplastic*?
- 7 What feature of the ethylene molecule allows it to link with other ethylene molecules?
- 8 Give two properties of LDPE that make it suitable for making shopping bags.
- 9 Why do some people want to reduce the numbers of polyethylene bags used?

Polyvinylchloride (PVC) This polymer is also an addition polymer. The monomer is monochloroethylene (vinyl chloride). PVC is strong and weather resistant. It is used for pipes, house cladding and enclosures for electronic equipment. It can also be treated to form a softer substance that is used in shrink wrapping and food packaging.

- 10 Give the molecular formula for vinyl chloride (see figure 2).
- 11 Give the molecular formula for PVC. (Hint: see question 3.)

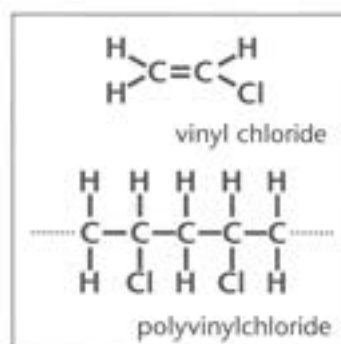


Figure 2

Polypropylene This polymer is similar in structure to polyethylene. It is also inexpensive but is stronger than polyethylene. It is used in making bottles, ropes, carpets and car parts. Figure 3 shows a short length of a polypropylene molecule.

- 12 Why is only part of the polymer shown?
- 13 What is the name of the monomer in polypropylene?
- 14 Draw the structural formula of the monomer in polypropylene.
- 15 Give the molecular formula of the monomer in polypropylene.
- 16 Is propylene likely to be *thermoplastic* or *thermosetting*? Explain your answer.
- 17 Petrochemicals are the raw materials used to make many plastics. What are *petrochemicals*?

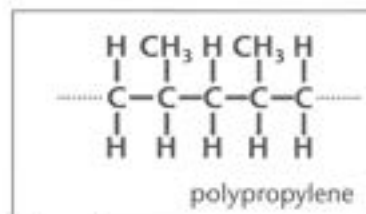


Figure 3

Polystyrene This is a rigid thermoplastic. Like many plastics, it is an excellent electrical insulator and is resistant to attack by chemicals. It is used to make disposable knives and forks, model kits and housings for large household appliances. It can also be 'foamed' to form styrene foam or Styrofoam™ which is much less dense than most materials. In this form it is used to make insulation for food and beverage containers, the 'beans' in 'bean bags' and protective packaging. Figure 4 gives the structure of the monomer in polystyrene: styrene (ethenylbenzene).

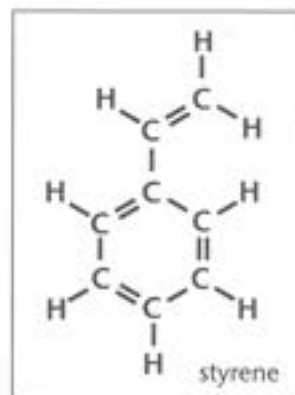


Figure 4

18 Give the molecular formula of styrene.

19 Why is expanded polystyrene a good material for making swimming aids and toys designed to float?

Polytetrafluoroethylene (PTFE) This polymer has the trade names 'TEFAL™' and 'TEFLON™'. The monomer is tetrafluoroethylene and has the formula C_2F_4 . PTFE is slippery and tough; it resists most chemicals and can withstand a wide range of temperatures. It is used in gaskets, sealants and protective coatings. The most common domestic use of PTFE is as the 'non-stick' coating on cooking utensils.

20 Draw the structure of tetrafluoroethylene. (Hint: there is a carbon-carbon double bond.)

21 Draw a section of the PTFE molecule.

Synthetic rubber There have been many types of synthetic rubbers produced; 'Buna' rubber was among the first. This polymer is not made up of one repeating monomer (**homopolymer**) but has two different alternating monomers (**copolymer**). Buna rubber has butadiene and styrene as monomers. It is named after *butadiene* and sodium, *Na*, which is a catalyst in its formation. World War II caused a shortage of natural rubber, which played a major role in the development of synthetic rubbers. After the war the Germans focused their research on producing a synthetic rubber that was better quality than the existing methyl rubber. Buna rubber was the result of their research.

22 Explain the difference between a *homopolymer* and a *copolymer*.

23 Buna rubber is made using butadiene, the simplest of the dienes. Define *diene*.

Nylon Nylon was originally a tradename but it is now a generic name for a range of polyamides. A polyamide has repeating amide ($R-CO-NH-R'$) groups. R and R' refer to different hydrocarbon groups. The two earliest nylons were Nylon 6 and Nylon 6, 6. Both of these are still very important commercially today. Figure 5 shows the similarity between the structures of these two polymers. Nylon is strong, wear-resistant and can be dyed. It is used to make clothing, carpets, parachutes, ropes and machinery parts. Nylon is a **condensation polymer**. Addition polymers have carbon-carbon double bonds in monomers, which break open and join with neighbouring monomers. Condensation polymers form when an -H 'ending' and an -OH molecule 'ending' from adjacent monomers are removed when the monomers combine. When monomers bond in condensation polymerisation, molecules of water are formed as well as the polymer chain.

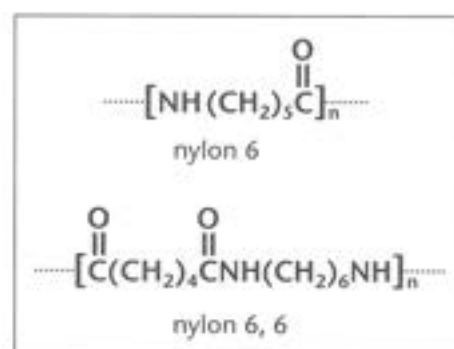


Figure 5

24 What is meant by a *generic name*?

25 Name the elements present in a polyamide.

26 Outline the major differences in bonding between an addition polymer and a condensation polymer.

27 Give one reason for the invention of synthetic rubber.

28 Draw up a table showing the names and uses of the plastics mentioned on these pages.

Plastics have been described as being 'too good'. They have become popular because they:

- are made from relatively cheap raw materials (petrochemicals)
- can be moulded and shaped or drawn into fibres
- are tough
- can be coloured
- are good thermal and electrical insulators
- have good safety and hygiene properties for packaging food and medicines
- are lighter than alternatives
- are resistant to chemicals.

The world's annual consumption of plastic materials has increased from around 5 million tonnes in the 1950s to nearly 100 million tonnes today. Plastics degrade very slowly. No one knows how long they take to break down because plastics have only been in existence for a relatively short time. Degrading slowly is a useful property while plastics are being used, but a problem when plastics are discarded. The disposal of plastics causes problems; e.g. they can give off toxic fumes when burnt. Many people think that less plastic should be used. Other people claim that plastics are an essential part of modern living; e.g. the use of plastics has made cars lighter and more fuel-efficient. There are ways that the amount of plastic thrown away can be reduced.

- Use less plastic; e.g. use alternative (conventional) products
- Recycle
- Develop biodegradable plastics.

The standard supermarket carrier bag is made out of LDPE. One estimate is that over 200 bags per person are given out and discarded in Australia every year. Many retail businesses are trying to discourage customers from using plastic bags by charging for them and by suggesting alternative ways of carrying goods.

- 1 What does the acronym *LDPE* stand for?
- 2 Why would a retailer want to reduce the number of plastic bags used by their customers?
- 3 At one supermarket, they decide to put customers goods in large paper bags instead of the usual plastic bags. Outline the possible advantages and disadvantages of this decision.
- 4 At another supermarket, a bin is provided in which people can put their old plastic bags to be recycled later. Outline the possible advantages and disadvantages of this.

'Green' plastics There are efforts to increase the amount of plastic being recycled, but there are problems with recycling plastic. Thermoplastics can be melted and reused but thermoset plastics cannot. The many different types have to be separated before recycling can begin, and this is very labour intensive. In addition, the value of the material is low so recycling plastics is not very profitable. Some vehicle manufacturers are modifying their designs to make the plastic parts easier to recycle when the vehicle reaches the end of its useful life. It is estimated that making plastic carrier bags from recycled material uses 35% of the energy and 10% of the water required to make bags from newly-manufactured polyethylene. It also produces about 50% of the pollution.

- 5 Clothes often come in clear plastic bags marked: *Not a toy. Keep away from children under the age of three.* Why do the plastic bags carry this warning?
- 6 Outline the advantages and disadvantages of making plastic carrier bags from recycled rather than 'virgin' polyurethane.

Scientists have developed polymers that break down. These can be **biodegradable** or **photodegradable**. Biodegradable plastics will degrade in landfill but can release methane – a greenhouse gas. Photodegradable plastics only break down on exposure to light.

Biopolymers have been developed from renewable resources. Genetically-modified organisms such as plant sugars can be used to make plastics. These plastics are still expensive and in the early stages of development. Some of the more important biopolymers are starch-based. This is because starch is easy to obtain, affordable, renewable and inexpensive to process as a biopolymer.

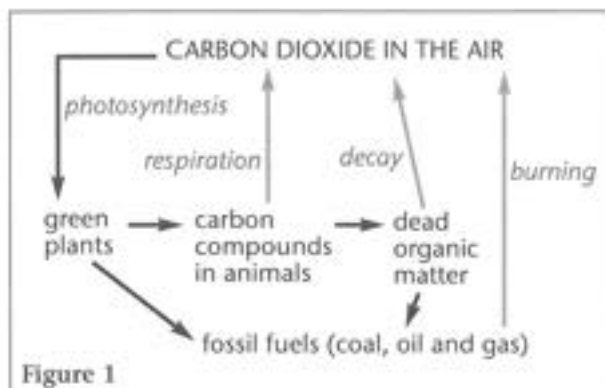
- 7 Will photodegradable plastics break down in landfill? Explain your answer.
- 8 Outline the differences between a *biodegradable polymer* and a *biopolymer*.
- 9 Outline why plastics have become popular.
- 10 Name a naturally-occurring substance that can be used to make a biopolymer.

Humans are sometimes described as **carbon-based life forms**. Carbon dioxide (CO_2) is an important way of transferring carbon compounds between living things and their environment. Carbon is one element that is constantly being removed from compounds that are being broken down to be used to make new compounds. This process can be described as the **carbon cycle**. Part of the carbon cycle is shown in figure 1.

- 1 List the processes that add CO_2 to the atmosphere.
- 2 List the processes that remove CO_2 from the atmosphere.

Photosynthesis takes place in the leaves of a plant. This process requires heat and light energy from sunlight. Chlorophyll must also be present.

- 3 Define *photosynthesis*.
- 4 What role does chlorophyll play in photosynthesis?
- 5 What raw materials are needed for photosynthesis?



Inside a plant the simple glucose molecules ($\text{C}_6\text{H}_{12}\text{O}_6$) can be joined together to form starch, cellulose and other complex carbon compounds. Animals then obtain carbon compounds (and energy) by eating plants. Humans obtain carbon compounds by eating both animals and plants. All food on Earth ultimately derives its energy from the Sun. Only about 1% of the solar energy that hits a green plant is converted to stored (chemical) energy. If these plants are not eaten directly by humans but used as fodder for animals used for meat, then more energy is 'lost' to us because some energy is used up by the animal. About one-tenth of the energy in the plants eaten by cattle is converted into stored energy (i.e. meat).

- 6 What percentage of solar energy is turned into stored energy in an animal used for meat?

If plants are allowed to rot and then compressed for a long period of time (millions of years), the carbon compounds eventually change to coal. Crude oil (petroleum) and natural gas (methane) are formed in a similar way from other organic compounds.

- 7 Approximately how long does coal take to form?

Respiration is the process by which plants and animals give out carbon dioxide. It is the reaction that living things use to get the energy they need. Respiration releases as heat, the energy (originally from sunlight) that is stored as chemical energy in glucose. Respiration takes place in cells. The reaction can be written as: $\text{glucose} + \text{oxygen} \rightarrow \text{carbon dioxide} + \text{water} + \text{energy}$.

- 8 Write a balanced equation for respiration.
- 9 Photosynthesis is an exothermic reaction. What is meant by *exothermic*?

Bacteria and other decomposers feed on dead plants and animals. Carbon dioxide is produced at the same time. Burning fuels also produces carbon dioxide.

- 10 Name three common fuels.

Carbon dioxide contributes to the **Greenhouse effect**. This effect keeps in heat and helps keep the Earth within a steady temperature range. Many people are concerned that we are burning too much fossil fuel and that this will increase the amount of carbon dioxide in the atmosphere.

- 11 What could too much carbon dioxide in the atmosphere do?
- 12 Name two substances that plants make using glucose.
- 13 What is the role of decomposers in the carbon cycle?
- 14 Mark each statement as true or false.
 - a Respiration describes the synthesis of complex organic compounds.
 - b Photosynthesis is a heat-absorbing process.
 - c Glucose is the base 'ingredient' for starch formation.
 - d All living things contain carbon.

Research 15 Oceans have not been mentioned on this page. Outline how oceans (and the organisms living in them) contribute to the level of carbon dioxide found in the air.

Carbohydrates are the most common class of organic (carbon-based) compounds present in organisms. Their name comes from *carbon hydrates* because many of carbohydrates have a formula that can be written as $C_x(H_2O)_y$. Many carbohydrates are formed by the process of **photosynthesis**. In photosynthesis, sunlight provides the energy to change carbon dioxide and water to carbohydrates in the presence of chlorophyll. The most common carbohydrate is **glucose** ($C_6H_{12}O_6$).

- Copy and balance the following reaction to show the formation of glucose by photosynthesis.
 $CO_2(g) + H_2O \rightarrow C_6H_{12}O_6$
- Is photosynthesis an *exothermic* reaction (heat given out) or *endothermic* reaction (heat taken in)? Explain your answer.

Glucose is a monosaccharide; it can't be broken down into two or more simpler sugars. Glucose is a hexose – there are 6 carbon atoms per molecule. Fructose is a similar molecule. A molecule of glucose can join with a molecule of fructose to form the disaccharide, sucrose. A molecule of water is also formed in this process. Figure 1 shows how glucose and fructose combine to form sucrose.

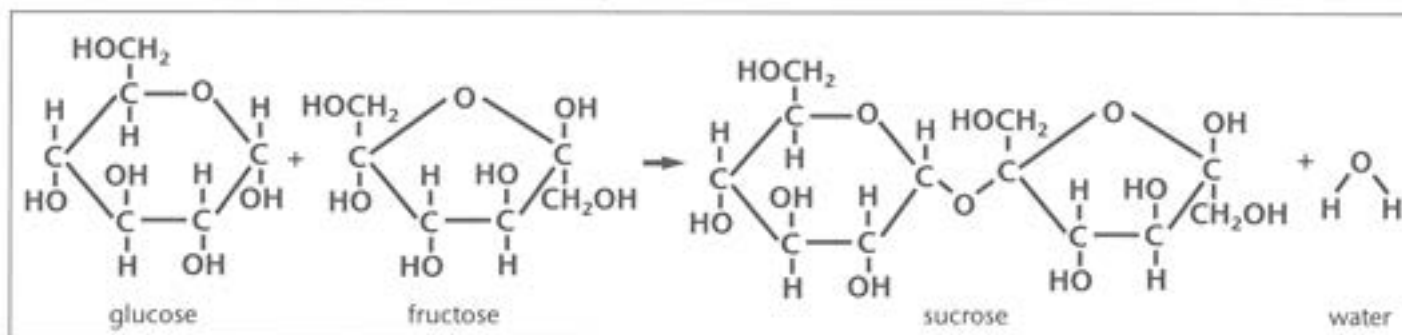


Figure 1

- Copy figure 1.
- Use figure 1 to help you write a balanced equation for the formation of sucrose.
- Identify one difference between a glucose molecule and a fructose molecule.
- Explain why sucrose is called a *disaccharide*.
- The formation of sucrose is a condensation reaction. What is meant by *condensation*?
- What is a *hexose*?

Polysaccharides are formed by joining many monosaccharides together. Cellulose, starch and glycogen are all polysaccharides that use glucose as the monomer (basic 'building block'). Cellulose has the formula $(C_6H_{11}O_5)_n$, where n can be from around 500 to 5000. Cellulose is being investigated as a replacement for petroleum hydrocarbons, because it is a renewable resource (being a major component of plants). Cellulose is not soluble in water. Starch is also derived from plants and insoluble in cold water but the glucose units are bonded in a different way. Approximately 3000 glucose units will make up a typical starch molecule. The test for starch is that it gives a blue/black colour when added to iodine solution. Glycogen (animal starch) has a more branched structure than starch and it is soluble in water. Glycogen is used by vertebrates for food storage.

- List the similarities and differences between cellulose, starch and glycogen.
- What is the chemical test for starch?
- Why is cellulose being considered as a replacement for petroleum hydrocarbons?

Reducing sugars have a free group for chemical bonding. They react with copper (II) ions (which are blue) to copper (I) compounds (which are red). They are called 'reducing' sugars because they can reduce copper (II) to copper (I); i.e. Cu^{2+} to Cu^+ . Benedict's solution is the most common reagent used. (A **reagent** is a substance used to measure or detect the presence of other substances.) Glucose, fructose, maltose (malt sugar) and lactose (milk sugar) are reducing sugars. Sucrose is not a reducing sugar.

- Which ions in Benedict's solution are responsible for identifying reducing sugars?

Proteins are needed in the body for the growth and repair of tissues and to help certain chemical reactions take place. They are used as an energy source only under extreme conditions. (Energy is the capacity to do work.)

- 1 Name one food group that is a major source of energy.

Proteins are very complex molecules. They are organic compounds that are used to make enzymes (biological catalysts) and hormones in the body. A hormone is a chemical produced by a gland and then carried in the bloodstream to regulate body functions. Hormones act as chemical messengers sent to produce a specific reaction. Proteins are **polymers** (very large molecules) that are made up of smaller molecules called **amino acids**. Amino acids contain both the -NH_2 (amine) and -COOH (carboxylic acid) groups.

Natural amino acids have quite complex structures so are often referred to by their 'trivial' names rather than by IUPAC nomenclature. They can be represented by the general formula $\text{H}_2\text{N-CH(Z)-COOH}$. (See figure 1.) The 'Z' represents the side chain and is responsible for the differences between amino acids. In α -amino acids, the carboxyl, amino and side chain are all attached to one central carbon atom. In aqueous solution the amino group can act as a base and the carboxyl group as an acid.

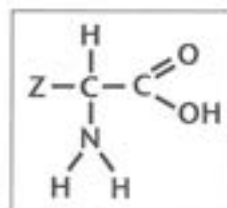


Figure 1

- 2 The carboxyl and amino groups are very polar. How will this affect the solubility of amino acids in water?
- 3 Outline how amino acids were named.
- 4 Glycine is the simplest amino acid. Its side chain ('Z') is hydrogen. Draw the structural formula of glycine.
- 5 Alanine has the side chain -CH_3 . Draw the structural formula of alanine.

Amino acids can join together to form longer molecules. If the chain is up to 10 amino acids long, it is sometimes called a **peptide**.

- 6 What do you think a *polypeptide* is?

In living things there are about 20 different α -amino acids. Proteins make up about 50% of a human body's dry weight. They are found in all body cells and have these functions:

- Make up muscles and tendons
- Give the skin its elasticity and strength
- Reinforce bones and teeth
- Make up antibodies, which are part of the body's immune system
- Make up haemoglobin, which is part of the blood's transport system
- Make up enzymes and hormones, which control chemical reactions in the body.

The name **protein** is derived from the Greek *proteios*, meaning 'primary' or 'of the first rank'.

- 7 Why do children need a greater percentage of proteins in their diet than adults do?

Proteins are formed when amino acids join together by condensation reactions. In these reactions, a carbonyl to nitrogen bond is created. An -OH group from one molecule and a -H from another molecule form a separate molecule of water. This is called a **peptide bond**.

- 8 Is this carbon to nitrogen bond covalent or ionic? Give a reason for your answer.

Figure 2 shows how two simple amino acids, glycine and alanine, can react to form a dipeptide.

- 9 Show another way in which glycine and alanine can join. (Hint: reverse the positions of the two amino acids as shown in figure 2).

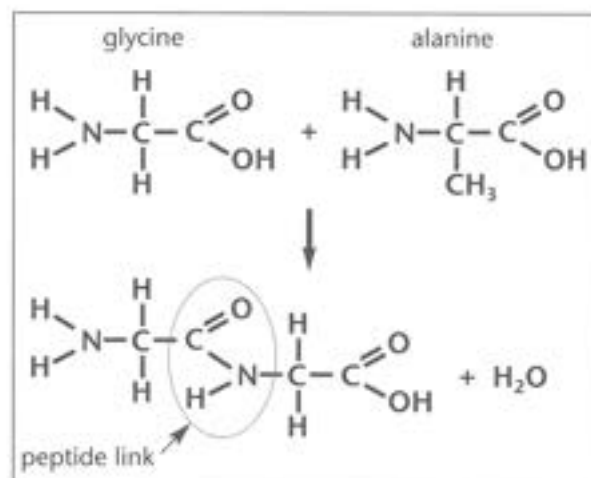


Figure 2

Protein structure

How proteins behave and what they do depends upon their shape. Proteins have **primary**, **secondary** and **tertiary** structures.

Primary structure is the order in which the amino acids are arranged; e.g. alanine, followed by glycine then lysine (ala - gly - lys) and so on. In other words, primary structure is the building of the initial chain.

Secondary structure is when the chain coils or folds to form a three-dimensional shape. This is because parts of the chain can attract each other. The two most common shapes are a flat sheet and a helix (helical). In other words, the secondary structure is the twisting of the chain that was formed into its given shape.

10 Draw the shape of a helix. (You may need to refer to a textbook.)

The attractions between the parts of the protein chain are mainly due to hydrogen bonds between $-C=O$ and $-NH_2$ groups in neighbouring amino acids.

11 What is a $-C=O$ group called?

12 What is a $-NH_2$ group called?

Hydrogen bonds are the weak attractions between dipoles (slightly-charged areas) existing in polar molecules. Figure 3 shows the hydrogen bonding between oxygen (δ^- , slightly negative) and hydrogen (δ^+ , slightly positive).

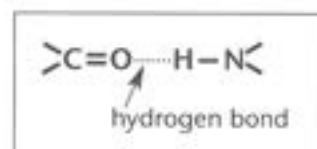


Figure 3

13 Copy figure 3 and show the relative charges on the atoms.

Tertiary structure is when distortions occur in the three-dimensional shape of a protein. These distortions are caused by the large Z-groups of some amino acids. There can also be hydrogen bonding between positive ($-NH_3^+$) and negative ($-COO^-$) groups. In other words, the tertiary structure is a further shaping of the secondary structure.

14 Outline the differences between the primary, secondary and tertiary structures of proteins.

The formation of a peptide link between amino acids can be reversed. Proteins can be hydrolysed (broken down) by heating with aqueous acids. In digestion, proteins are hydrolysed much more easily by using **digestive enzymes** such as pepsin and trypsin. **Hydrolysis** (reaction with water/ OH^-/H^+) can be used to identify the amino acids present in a protein. A protein can be hydrolysed and the products used in paper chromatography to separate and identify the components.

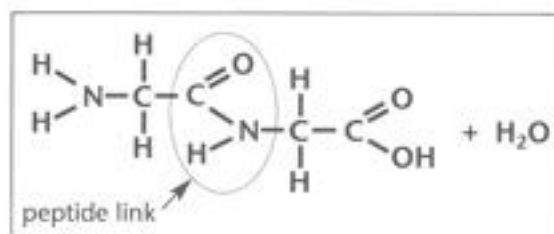


Figure 4

15 Copy figure 4 and show the products of this reaction.

Enzymes Enzymes are proteins that control the reactions which take place in the body. Enzymes have some similarities with inorganic catalysts (i.e. they speed up reactions) but there are important differences. They are more specific than inorganic catalysts. Enzymes do not catalyse lots of reactions, but usually specialise in producing one reaction only. Enzymes work under milder conditions and tend to operate at body temperatures instead of the high temperatures and pressures needed in many industrial processes.

The secondary and tertiary structures give each enzyme a unique cavity or pocket called an **active site**. This active site will only hold molecules of a specific shape; this is why it is often called a **lock-and-key** structure. The sequence of an enzyme-catalysed reaction is:

- The reactant (substrate) goes into the active site
- Bonds are formed between the substrate and the enzyme
- Bonds are broken within the substrate
- The substrate breaks or rearranges into new products
- The products are released from the active site.



Figure 5

16 Copy figure 5 and label the following: *enzyme*, *active site*, *substrate*, *product*.

17 Outline the similarities and differences between inorganic catalysts and enzymes.

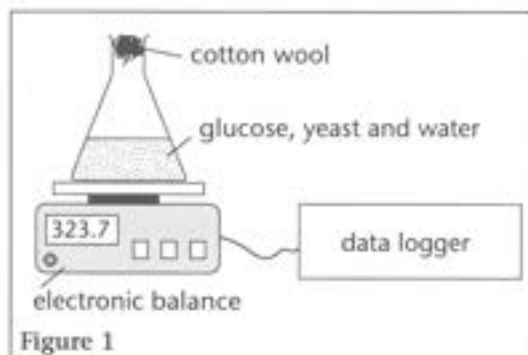
When yeast is added to a solution of glucose ($C_6H_{12}O_6$), it can ferment the sugar solution to ethanol (C_2H_5OH) and carbon dioxide (CO_2). The ethanol remains in solution and the carbon dioxide is given off as a gas.

- 1 Write a balanced formula equation for the fermentation of glucose solution.

A student wanted to investigate the rate at which glucose was changed to ethanol by yeast. Below is part of his practical report.

Aim: To investigate the progress of a fermentation reaction by measuring mass changes

Method: 10.0 g of glucose was dissolved in 200 mL of water. A pinch of yeast (approx. 0.1 g) was added and the mixture was thoroughly shaken. The top of the conical flask was loosely plugged with cotton wool. (This was to allow carbon dioxide to escape, but prevent microbes from entering.) The flask and contents were placed on an electronic balance and data logger as shown in figure 1. The data logger was set to record the mass of the flask and contents each hour for 5 days.



Results:

Mass of empty flask = 113.6 g

Mass of flask + water + glucose + yeast (at time 0) = 323.7 g

Sample readings:

Time (hours)	0	5	10	15	20	25	30	35	40	45	50	55	60
Mass (grams)	323.7	322.5	321.3	320.5	319.9	319.6	319.4	319.3	319.3	319.3	319.3	313.3	313.3

- 2 Explain why the mass of the flask and contents decreased over time.
- 3 Why did the student use a data logger rather than take the readings manually?
- 4 The sample readings do not show all of the readings. Why?
- 5 Draw up a table that shows time (hours) and total mass lost (grams).
- 6 Draw a graph to show the relationship between the mass lost (y-axis) and time (x-axis).
- 7 What is the dependent variable in the experiment?
- 8 What is the independent variable in the experiment?
- 9 When was the reaction complete? Explain your answer.
- 10 What mass of glucose was added?
- 11 What is the mass of one mole of glucose? Show your working. (Hint: C = 12.01, H = 1.008, O = 16.00)
- 12 How many moles of glucose were added? Show your working.
- 13 What mass of carbon dioxide evolved?
- 14 What is the mass of one mole of carbon dioxide?
- 15 How many moles of carbon dioxide were evolved? Show your working.
- 16 Assuming that one mole of carbon dioxide occupies 24 litres, what was the volume of carbon dioxide generated? Show your working.
- 17 After carrying out the experiment, the student wondered how changing the temperature would affect the rate of fermentation. Outline how he could carry out investigations to find out. Include details of how he could ensure that the results of the investigation were valid and reliable.
- 18 Define *fermentation*.
- 19 Give three examples of ways fermentation is used and what is produced.

Ethanol (ethyl alcohol, $\text{CH}_3\text{CH}_2\text{OH}$) is a clear, colourless liquid. It is miscible (mixes) with water and burns forming carbon dioxide and water.

- 1 Write a balanced formula equation for the combustion of ethanol.

Ethanol can be added to petrol (up to 10%) to form **gasohol**. Ethanol is a very important solvent and a raw material for other industrial chemicals.

Ethene (ethylene, CH_2CH_2) is a flammable, colourless gas that is insoluble in water.

- 2 Figure 1 shows the structural formulae of ethanol and ethene. Copy figure 1, name each molecule and give its molecular formula.

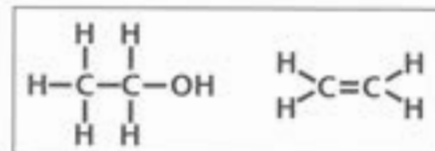


Figure 1

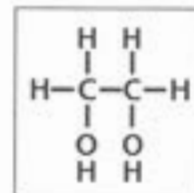


Figure 2

Ethene is used to make polythene, ethanol and 1, 2-ethanediol (ethylene glycol, 'antifreeze'). Figure 2 shows the structural formula of 1, 2-ethanediol.

- 3 Copy figure 2. Name the molecule and give the molecular formula.

Ethanol is made industrially by two processes: anaerobic (no oxygen present) fermentation of carbohydrates and by the hydration (addition of water) of ethene. All ethanol used in beverages (drinks) is obtained by fermentation. Raw materials used for fermentation include grains, potato and sugar. Fermentation is an enzyme-controlled reaction. In the presence of yeast, glucose ($\text{C}_6\text{H}_{12}\text{O}_6$) is fermented to ethanol and carbon dioxide gas. The yeast contains enzymes (organic catalysts) that enable the reaction to take place at room temperature. The maximum ethanol concentration obtained by fermentation is around 15% – any higher than this and the yeast is poisoned by the ethanol. Ethanol can be obtained from an ethanol/water mixture by fractional distillation.

- 4 Write a balanced formula equation for the fermentation of glucose solution to ethanol and carbon dioxide.

Most industrial ethanol is made from ethene because it is cheaper to make it this way. The process involves the acid-catalysed hydration of ethene. The catalyst is phosphoric acid (H_3PO_4).

- 5 Write a balanced formula equation for the hydration of ethene to ethanol.

Ethene is produced by the catalytic cracking (breaking up) of hydrocarbons obtained from petroleum (crude oil). Figure 3 shows how ethene and hydrogen can be produced by the cracking of ethane.

- 6 Copy figure 3 and write a balanced formula equation for this reaction.

- 7 Give three uses of ethanol.

- 8 Give three uses of ethene.

- 9 Name the raw materials used in the formation of ethanol by fermentation.

- 10 Explain what is meant by the each term.

- a Miscible
- b Gasohol
- c Anaerobic
- d Hydration
- e Enzyme
- f Cracking
- g Fermentation

- 11 Give the formula for phosphoric acid.

- 12 Why is there a limit to the percentage of ethanol in a solution formed by fermentation?

- 13 Name the process used to extract ethanol from an aqueous solution.

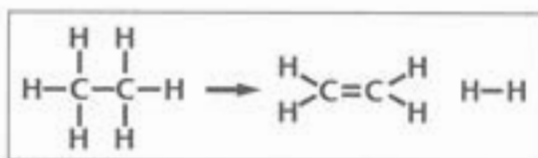


Figure 3

The various scents of fruits and flowers are caused by the presence of esters; so it should come as no surprise to learn that esters are often used as natural flavourings. However, esters are most commonly used as an important part of industrial solvents.

Esters have two carbon chains separated by an oxygen atom. They have the general formula $R-COOR'$, where R is derived from an organic acid and R' from an alkanol (alcohol). Figure 1 shows the general structure of an ester. Esters can be detected in water in concentrations as low as a few parts per billion (ppb). Esters can be produced by reacting an organic acid with an alkanol (alcohol) in an equilibrium reaction:



Figure 2 shows the general structure of an alkanol. Figure 3 shows the general structure of an alkanolic (carboxylic) acid. The simplest ester is formed from the reaction between methanoic ($HCOOH$, formic) acid and methanol (CH_3OH). It is called methyl methanoate (methyl formate). Figure 4 shows the structure of methyl methanoate.

Esterification is a condensation reaction; a molecule of water is formed. The reaction can be speeded up by adding concentrated sulfuric acid which removes the water produced and drives the equilibrium reaction to the right.

- Figure 5 shows the reaction between methanoic acid and methanol. Copy figure 5 and write a balanced equation for this reaction.

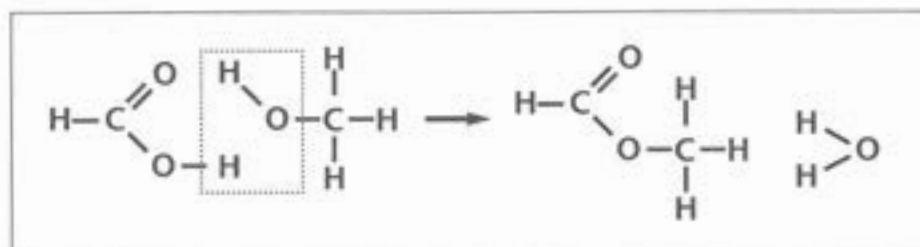


Figure 5

In the names of esters, the alkanol contributes the first part of the name and the acid the second part. For example, alkanolic acids form alkananoates, methanoic acids form methanoates, and ethanoic acids form ethanoates. Note that the name is the 'reverse' of the order in which the acid structure is usually drawn.

- Using figure 5 as a model, show the reaction between ethanol and ethanoic acid (CH_3CHOOH) forming ethyl ethanoate.
- Copy and complete the table. The first example has been done for you.

Name of ester	Formula of ester	Parent acid	Parent alkanol	Scent
Methyl butanoate	$C_4H_7OOCH_3$	Butanoic	Methanol	Pineapple
		Ethanoic	Pentanol	Banana
Ethyl methanoate				Raspberry
Octyl ethanoate				Orange

- What is represented by *ppb*?
- What is a *condensation* reaction?
- What is a *fragrance*?
- Do you think that esters play a role in the making of perfumes? Explain your answer.

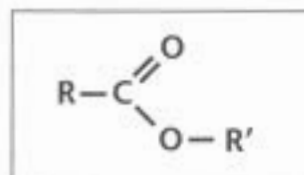


Figure 1

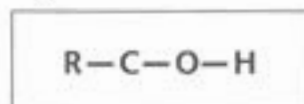


Figure 2

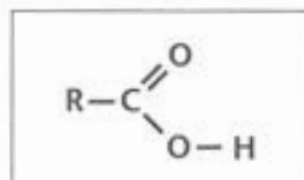


Figure 3

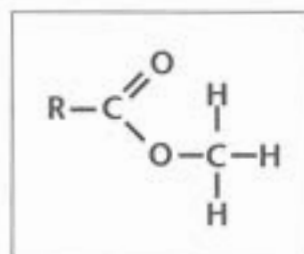


Figure 4

The atmosphere is the name for the air (gases) surrounding the Earth, reaching from the Earth's surface to the edge of space. Scientists have divided the atmosphere into various layers (usually based on temperature).

Figure 1 (not to scale) shows a simplified view of the various atmospheric layers. There are no distinct boundaries and overlapping between different layers does occur; this is why textbook values can vary slightly.

Some areas exhibit specific physical or chemical characteristics, but don't quite fit in the diagram shown. For example, the ozonosphere is a region where ozone is produced and the ionosphere contains charged particles (ions) that allow radio signals to be bounced from one country to another.

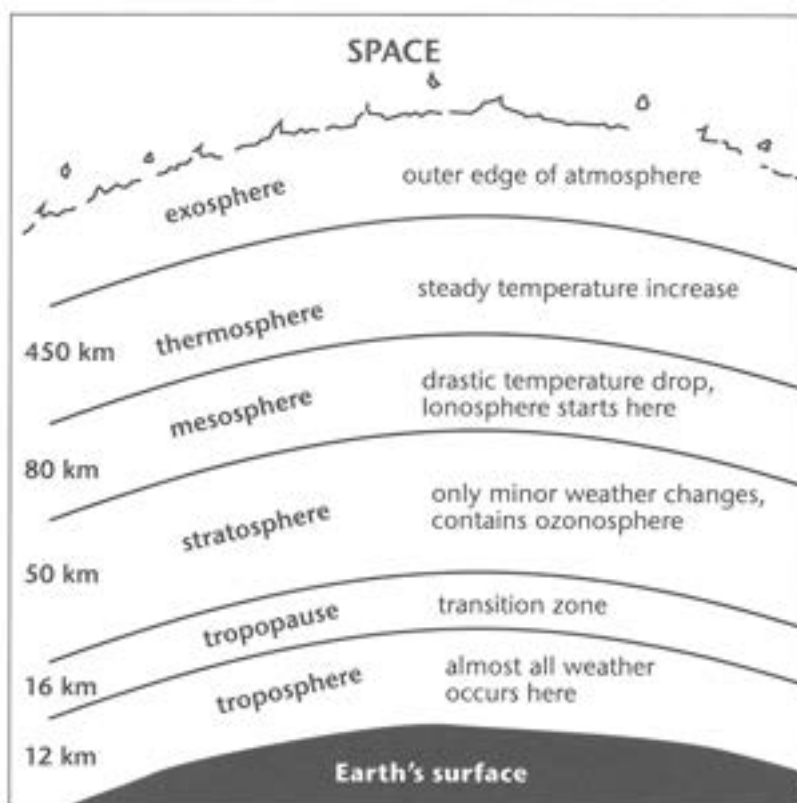


Figure 1

- 1 Draw up a table to summarise the information above and in figure 1 about the atmospheric layers. Use the headings shown below.

Name of layer	Distance from surface (km)	Features of layer

Air pressure A huge amount of air is pressing down on us at the Earth's surface. Although we cannot see air pressure, it can be measured. Air pressure varies with the height above ground and the weather, but a standard pressure at sea level is 1 atmosphere (equal to 101.3 kilopascals, 760 mm of mercury, 14.7 pounds per square inch and 1013 millibars).

- 2 The weather forecast includes air pressure readings along with temperatures and other features of the day's weather. Record your local air pressure readings, temperature and weather conditions for five consecutive days. Draw conclusions as to how, or if, connections can be made between air pressure, temperature and weather conditions.

Composition of the atmosphere The three main gases present in the atmosphere are nitrogen (N_2 , 78.08%), oxygen (20.95%) and argon (Ar, 0.94%). Water vapour (H_2O) is also present in variable amounts depending on weather conditions. The remaining gases, although present in minute amounts, still form an important part of the atmosphere. These include carbon dioxide (CO_2), neon (Ne), helium (He), methane (CH_4) and krypton (Kr). Other gases present in variable amounts include ozone (O_3) and sulfur dioxide (SO_2).

- 3 Draw up a table to summarise the information about atmospheric gases. Use the headings shown below.

Name of gas	Formula	Element or compound	Amount in air (%)

- 4 If you were to ascend through the various atmospheric layers, would you be able to see distinct boundaries? Explain your answer.

Research 5 Methane and carbon dioxide are known as *Greenhouse gases*. Find out what is meant by this term and why the word *greenhouse* is used.

Separation of gases in the atmosphere Air is a source for many gases used in the chemical industry. Gases can change to liquids through a process of compressing and cooling. The difference in boiling points of gases can be used to separate out the different gases in air. In this process, air is liquefied and then gradually allowed to warm up. Each component changes back to a gas when it reaches its boiling point. (Table 1 shows the boiling points of some gases.) Each gas is extracted as it evaporates out of the liquid at its specific boiling point. This process of **fractional distillation** does not give pure gases initially; further stages are required.

6 Two common ways of describing temperature are the Celsius ($^{\circ}\text{C}$) and Kelvin (K) scales. Explain how to convert degrees Celsius into Kelvins.

7 Consider a liquid sample of air that has been cooled to -200°C . If the only gases present are nitrogen, oxygen and argon, in what order would they turn to gas as the mixture warmed?

Gas	Boiling point ($^{\circ}\text{C}$)	Boiling point (K)
Nitrogen	-196	77
Oxygen	-183	90
Argon	-186	87

Table 1

Nitrogen and its compounds Nitrogen has no taste, no smell, no colour and is relatively unreactive. It exists as diatomic molecules.

8 Explain what is meant by a *diatomic molecule*.

Nitrogen was first isolated around 1772 by several scientists independently. These were Daniel Rutherford (Scotland), Joseph Priestley (England), Henry Cavendish (England) and Carl Wilhelm Scheele (Sweden). Antoine Lavoisier (France) was the first person to realise that nitrogen was an element.

Research 9 Find out one more fact about each of the scientists mentioned above.

Nitrogen has many uses that rely on the fact that it doesn't support oxidation or combustion. Nitrogen is used to replace air in the packaging of some foods (to prevent oxidation). In its liquid form it is used to 'flash freeze' foods. In this process liquid nitrogen is sprayed onto foods and they freeze almost immediately. The nitrogen then evaporates. The frozen food, on thawing, retains most of the properties of fresh food because the freezing process is so quick. Liquid nitrogen is also used to preserve biological samples.

10 A manufacturer of premium potato chips filled each packet with nitrogen gas (as well as chips!) before sealing the packet. Why?

11 Give two uses of liquid nitrogen.

Nitrogen compounds are a major factor in maintaining life. Nitrogen is an important component of amino acids (which form proteins) as well as the giant molecules DNA and RNA, the genetic material that carries the 'blueprint' information for living things. Nitrogen has to be returned to the soil after crops have been harvested. A major use of nitrogen is in the Haber process. This process produces ammonia (NH_3) from hydrogen and nitrogen. Ammonia is used to make nitric acid; this in turn is the raw material in producing nitrates for use in dyes, explosives and fertilisers. Certain species of bacteria live either separately or associated with certain plants (legumes) and are known for **nitrogen fixation**.

12 Write a balanced formula equation for the formation of ammonia from nitrogen and hydrogen.

Research 13 Use reference materials to give a brief definition for *nitrogen fixation*.

14 Name one legume.

Oxygen This gas was discovered by Joseph Priestley in 1774. He heated mercury oxide (HgO), forming mercury and oxygen. Oxygen was further investigated by Antoine Lavoisier who discovered many of its properties and gave oxygen its name.

15 Write a balanced formula equation for the thermal decomposition of mercury oxide.

Oxygen is similar to nitrogen in that it has no colour, taste or odour. It is a very reactive element and is needed by animals for respiration. The greatest industrial use of oxygen is in blast furnaces producing steel. Other uses include life support for hospital patients and oxy-acetylene welding. Acetylene (ethyne) has the formula C_2H_2 and when burned with oxygen produces carbon dioxide and water. Until 1961, oxygen was used as the standard for atomic masses. The isotope oxygen-16 was assigned a mass of 16.000 amu.

16 Write a balanced formula equation for the burning of acetylene in oxygen.

Research 17 Which isotope do scientists now use as the standard for atomic masses?

The Noble gases In 1904 Sir William Ramsay was awarded the Nobel Prize in Chemistry for discovering a new Group of elements, the Noble gases. This Group is now known to consist of helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and radon (Rn). Radon was discovered by Friedrich Ernst Dorn in 1898. These gases are all colourless, tasteless and very unreactive. They are unreactive because they have full outer (valence) electron shells. In total, the Noble gases make up around 1% of the earth's atmosphere (by volume); argon is by far the most common. Although helium is estimated to make up around 23% of the material in the Universe, it only makes up about 0.0005% of Earth's atmosphere.

Helium (from the Greek word *Helios*, meaning Sun) is used in cryogenics because it can (when liquid) reach temperatures very close to absolute zero (-273.15°C). Cryogenics is the science that deals with the behaviour of materials at very low temperatures (*cryo*: icy; *genic*: produced by nature). Another use of helium is in lighter-than-air balloons.

When **neon** (from the Greek *neos*, meaning new) gas has an electric current passed through it while under low pressure, the gas emits a reddish light. This makes neon useful in making advertising signs and also in some lasers.

Argon (from the Greek *argon* meaning lazy) can be used when welding reactive metals to provide an inert atmosphere to stop the metals from reacting with oxygen. Light globes can also contain argon. In these globes a thin tungsten (W) wire glows white-hot when an electric current is passed through it. The inside of the globe is filled with argon.

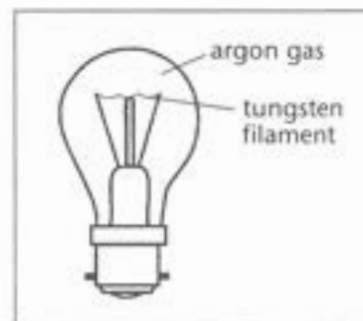


Figure 1

- 1 Give a reason why the gas argon is used instead of air in the light globe shown in figure 1.

Krypton (from the Greek *kryptos*, meaning hidden) forms some compounds with oxygen and chlorine. It is also used in lighting. **Xenon** (from the Greek *xenon*, meaning stranger) is the most reactive of the Noble gases. It is used in powerful lamps and radiation detectors. **Radon** is a radioactive gas that is produced by the decay of radium (Ra). Radon can be used to treat cancers (radiotherapy) and in taking X-ray type pictures of the human body (radiography).

- 2 The noble gases are all monatomic. Explain what is meant by *monatomic*.
- 3 The noble gases can be extracted from liquid air by *fractional distillation*. What property of the gases allows fractional distillation to occur?
- 4 Draw up a table to summarise the information about the noble gases. Use the headings shown below.

Name	Symbol	Use	Origin of name

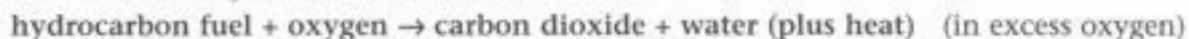
- 5 What is *cryogenics*?
- 6 Give the symbol for tungsten.
- 7 Give the symbol for radium.
- 8 Why are the Noble gases so unreactive?

Carbon dioxide This gas is naturally present in the atmosphere, but levels have been increasing since humans began burning fossil fuels (coal, oil and natural gas) in large quantities. It is a significant greenhouse gas that contributes to global warming. Many people are concerned about the possible consequences of increasing global warming. Carbon dioxide (CO_2) turns limewater (calcium hydroxide) solution, $\text{Ca}(\text{OH})_2$ (aq), 'milky' forming insoluble calcium carbonate (CaCO_3). This reaction is used as a test for carbon dioxide gas.

- 9 Write a balanced formula equation for the reaction between carbon dioxide and limewater.
- 10 Pollution can be defined as: *The contamination of an environment, especially with waste made by humans*. Do you consider carbon dioxide a pollutant? Explain your answer.
- 11 Mark each statement as true or false.
 - a The Noble gases form a large proportion of Earth's atmospheric gases.
 - b Carbon dioxide would not be present in the atmosphere if humans hadn't polluted it.
 - c Oxygen turns lime water from clear to milky.

Pollution in the atmosphere

There are substances in the atmosphere that we think are harmful to both the environment and people. These substances are often grouped under the title **pollution**. Sources of pollution include motor vehicles, industry, power-generating stations and human homes. The petrol (internal combustion) engine has the following theoretical reaction:



Octane (C_8H_{18}) is a liquid hydrocarbon often used as a benchmark for judging other fuels (petrol).

12 Define *pollution*.

13 Write a balanced formula equation for the combustion of octane in a plentiful supply of air.

Unfortunately, in practice, the reaction of hydrocarbon fuel and oxygen (even with a pure hydrocarbon fuel) doesn't quite follow this path of forming the products carbon dioxide and water. Other products include unburned hydrocarbons, carbon monoxide and nitrogen oxides. Sulfur dioxide and particulate matter (small particles that can pass into the lungs) are also produced by diesel engines.

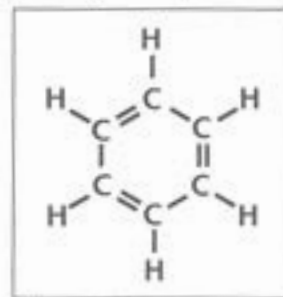


Figure 2

Hydrocarbons One hydrocarbon that causes special concern is benzene (see figure 2). Petrol and diesel fuel contain only about 1% benzene, but it is toxic and carcinogenic (cancer-forming).

14 What is the molecular formula for benzene?

15 Apart from exhaust fumes, how can hydrocarbon fuels be released into the atmosphere?

16 Give two reasons why contact with benzene should be minimised.

Carbon monoxide (CO) Wood stoves, cigarette smoke and unvented gas heaters are also sources of carbon monoxide. Carbon monoxide is a colourless, odourless and poisonous gas.

17 Explain why very high levels of carbon monoxide occur in areas of traffic congestion.

Nitrogen oxides NO_x is often used as a shorthand way of representing the various oxides of nitrogen. NO_x is produced by vehicles and power stations. It is harmful to breathing systems and combines with unburned hydrocarbons to form ground-level ozone (O₃) and smog. It is also a major contributor to acid rain.

18 Nitrogen dioxide (NO₂) can dissolve in water to form dilute nitric acid (HNO₃). Write a balanced formula equation for this reaction.

19 In addition to acid rain, name two problems that NO_x can cause.

Sulfur dioxide (SO₂) All fossil fuels contain some sulfur; sulfur dioxide is formed when fossil fuels burn. Power stations using coal and oil are the major human producers of sulfur dioxide. Volcanoes produce sulfur dioxide. When sulfur dioxide dissolves in water in the atmosphere it forms sulfurous acid (H₂SO₃) which is a contributor to acid rain.

20 Write a balanced formula equation for sulfur dioxide and water forming sulfurous acid.

21 Name a gas (apart from sulfur dioxide) that is a contributor to acid rain.

22 Briefly explain what acid rain is and how it is created/formed.

Particulate matter This term covers all of the tiny particles that can be suspended in the air for long periods of time. Dust, droplets of liquids, soot (carbon particles) and smoke are the major components of particulate matter in the air.

Ozone When this gas is in the upper areas of the atmosphere, it plays an important role in screening out harmful ultra violet (UV) rays. If it is at ground level, ozone irritates the eyes and damages the respiratory system.

23 Which element makes up ozone?

24 Outline one positive contribution of ozone.

25 Outline one negative contribution of ozone.

Research **26** Chlorofluorocarbons (CFCs) are considered a pollutant. Find out what they are used for and what harm they do.

27 Lead used to be added to petrol. Find out why it was used and why this practice has been discontinued.

28 *Pollution will always form a part of our lives; it is how we deal with pollution that matters.* Discuss.

You may need a textbook and/or dictionary to help you with this worksheet.

There are many roles that chemists play in modern society. Four main areas are:

- industry
- pure research
- education
- community work

1 Read the following passages and define each term in *italics*.

Chemists Three branches of chemistry are organic, inorganic and physical. Organic chemistry involves the study of carbon compounds; inorganic chemistry is the study of compounds not based on carbon; and physical chemistry is concerned with the physical characteristics of *matter*.

Industry In our world we depend on manufactured goods and there are many different industries that produce these goods. For example, chemists can be found working in the metal, *petrochemical*, *pharmaceutical*, agriculture, fibre, cosmetics, food and electronic industries. Industrial chemists tend to be employed in *quality control* and *research and development*. Chemists working in quality control ensure that the standard of goods produced is of the required quality. This means checking *raw materials* as they come into a factory and carrying out an *analysis* to make certain these materials are what they are supposed to be. Chemists need to check at many stages of the manufacturing process that conditions such as temperature, concentration of solutions and pressure are within specified *margins*. Chemists are also responsible for ensuring that factory *emissions* do not exceed specified limits. Research and development (R & D) is a process that improves existing *products* and manufacturing processes, and designs new ones. The properties of materials are investigated and these materials can be altered or used in new ways. The use of computers has speeded up much of the research done by chemists and many processes have been *automated*.

Pure research Sometimes chemists investigate substances and reactions without an obvious end product for *commercial* use. This pure research can lead to discoveries that are used by other scientists to produce goods or processes that help society. Pure research is often carried out by government bodies, some large firms and *academic institutions* because it is expensive, takes many years and is generally without immediate financial benefit.

Education *Chemistry* (either as a separate subject or as part of science) is taught in almost all high schools. Students who wish to study life sciences at tertiary level are often advised to take chemistry in the senior years of high school. Universities offer many graduate and postgraduate courses in chemistry and subjects like *biochemistry* where biology and chemistry overlap.

Community work Most chemists involved in community-related work are engaged in analytical and R & D roles. The major employers include:

- government research and scientific organisations, such as CSIRO
- local government
- hospitals and health care organisations
- law enforcement agencies.

Organisations like CSIRO often take on research tasks that may not seem immediately profitable, but have the long-term goal of contributing to public welfare. Local government may be concerned with environmental issues and hospitals with public health issues. *Forensic* chemists play a role in analysing evidence in criminal cases.

2 Quality control must be assessed throughout the whole production process and not just at the raw materials stage. Suggest why.

3 Why is pure research not carried out by many organisations?

4 What is the benefit of pure research if it is not known how the findings will be applied?

Research 5 Select one industry. Describe a role played by a chemist in that industry.

Sulfur (sometimes spelt *sulphur*) is a yellow, brittle solid at room temperature. It doesn't dissolve in water but is soluble in some organic (carbon-based) solvents such as methylbenzene. Being a typical non-metal, sulfur does not conduct electricity and has a low melting point (approx 115°C). Sulfur was called **brimstone** in old books. It occurs as the almost pure element in underground deposits (e.g. those found in Sicily and the USA). It is the sixteenth most-abundant element in the Earth's crust.

Sulfur is in Group VI in the Periodic Table and so sulfur atoms have six electrons in the outer (valence) shell. Figure 1 shows how the electrons are arranged in an atom of sulfur. Sulfur forms many compounds; e.g. oxides, sulfides, sulfites and sulfates. Sulfur dioxide (SO_2), iron (II) sulfide (FeS), sodium sulfite (Na_2SO_3) and copper sulfate (CuSO_4) are common examples of sulfur compounds.

The most important use of sulfur is in the manufacture of sulfur compounds such as sulfuric acid (H_2SO_4) which is a very important industrial chemical. It is important in medicine for '*sulfa*' (sulphonamide) drugs and in many skin treatments. Sulfur is also a raw material in the manufacture of matches, rubber tyres, dyes and gunpowder.

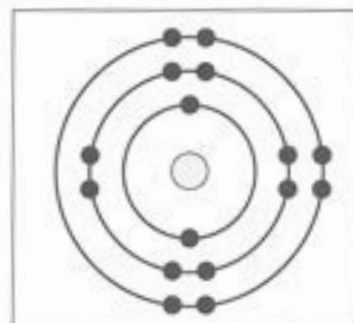


Figure 1

When sulfur burns in air, it gives a blue flame, forming sulfur dioxide. This is a colourless, heavier-than-air gas. Sulfur dioxide is also formed when fossil fuels such as coal and oil are burned. Coal usually contains 0.5%–5% sulfur. Sulfur dioxide has a very irritating smell and is poisonous even at low levels. It is thought to contribute to bronchitis and other lung diseases. Sulfur dioxide is acidic and has bleaching properties. This compound can be used to kill germs and, at low concentrations, as a food preservative, as it prevents the growth of moulds and bacteria.

Millions of tonnes of sulfur dioxide are emitted every year into the atmosphere by power stations. The sulfur dioxide reacts with water vapour, oxygen and forms sulfuric acid which then falls as **acid rain** – damaging buildings and killing trees and fish in lakes. Sulfur dioxide is also produced when some metals are extracted from certain ores; e.g. zinc from zinc blende (ZnS), iron from iron pyrites (FeS_2) and lead from galena (PbS). Large amounts of **quicklime** (calcium oxide, CaO) have been added to acidic lakes to neutralise the acidity.

World production of sulfuric acid is in the region of hundreds of millions of tonnes annually. The largest single use of sulfuric acid is in the manufacture of fertilisers. It is also used in making organic products, refining petroleum (crude oil), treating metals and making paints, pigments and artificial fabrics such as rayon. One of the more commonly known uses of sulfuric acid is in the car battery or **lead-acid accumulator**.

- 1 Give three characteristics of sulfur that indicate it is a non-metal.
- 2 Name another element that is in Group VI of the Periodic Table.
- 3 How many electrons does an atom of sulfur have in total?
- 4 In chemistry terms, what does *organic* mean?
- 5 Write the name and formula of each sulfur compound mentioned on this sheet.
- 6 What do you think the formula of sulfur trioxide is?
- 7 Give three products that use sulfur as a raw material in their production.
- 8 A label on a wine bottle said that a very small quantity of sulfur dioxide had been added. Why was this done?
- 9 Name one industry that is a major source of the sulfur dioxide in the atmosphere.
- 10 Name two chemicals present in a car battery.
- 11 Give two uses of sulfuric acid.

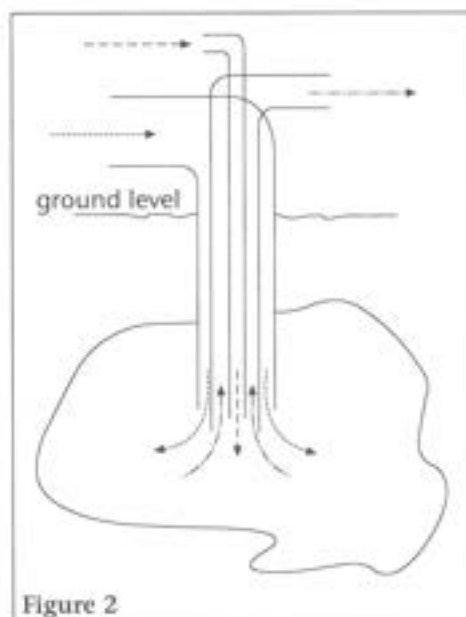
Research 12 Sulfur can exist as allotropes. What is an *allotrope*?

The Frasch process

Sulfur can be extracted as a by-product from refining petroleum and the processing of metal ores that contain sulfur. Most sulfur is obtained by an unconventional mining method called the Frasch process. This technique was invented in 1891 by Herman Frasch and relies on the low melting point of sulfur and its insolubility in water.

Figure 2 shows a simplified version of the mining operation.

- There are three pipes, stacked inside one another.
- These pipes are sunk into a deposit of sulfur.
- The outer pipe carries superheated steam (160°C) to the sulfur deposit.
- The sulfur is melted by the steam.
- The central pipe carries compressed air to the mixture of molten sulfur and steam.
- The compressed air forces the sulfur/water mixture to the surface via the intermediate pipe.
- On the surface, the mixture is collected and cools down. The sulfur solidifies and separates out.



13 Copy figure 2. Use the information above to label your diagram.

14 Which physical properties of sulfur allow the Frasch process to work?

The Contact process

Sulfuric acid is one of the world's most important chemicals and over 50 million tonnes is produced annually. About two-thirds of the sulfuric acid produced is used in the production of fertilisers. It is also used in the production of dyes and explosives and in the petrochemical industry. The Contact process is the major industrial method of producing sulfuric acid. The first stage in the Contact process is to produce sulfur dioxide (SO_2). This can be done by burning sulfur in air or by roasting metal sulfides.

15 Write a balanced equation for the formation of sulfur dioxide by the combustion of sulfur.

16 Write a balanced equation for the formation of sulfur dioxide by the reaction between oxygen and iron (II) sulfide (FeS) also forming iron (III) oxide (Fe_2O_3).

The next step is to oxidise sulfur dioxide to sulfur trioxide (SO_3) using vanadium (V) oxide as the catalyst at about 450°C .

17 Write a balanced equation for the formation of sulfur trioxide by the reaction between sulfur dioxide and oxygen.

Sulfur trioxide gas is passed through concentrated sulfuric acid (H_2SO_4) and an unstable substance called oleum ($\text{H}_2\text{S}_2\text{O}_7$) is formed (*oleo*, meaning oil). When water is added to oleum, sulfuric acid is formed.

18 Write a balanced equation for the formation of oleum by the reaction between sulfur trioxide and concentrated sulfuric acid.

19 Write a balanced equation for the reaction between water and oleum.

20 The chemists in charge of plants where the Contact process is carried out must ensure that sulfur dioxide does not escape into the atmosphere. Give a reason for this.

21 Name the major use of sulfuric acid.

22 Draw up a flow chart to show the Contact process.

23 Mark each statement as true or false.

- Hydrochloric acid is produced by the Contact process.
- Sulfur is soluble in water.
- Most sulfuric acid is used in the production of fertilisers.
- Sulfur is a by-product when petroleum is refined.

With the exception of certain meteorites, iron does not occur native (uncombined with other elements) in the Earth's crust. There are many compounds of iron; these include iron sulfide (iron pyrites, FeS_2), haematite (Fe_2O_3) and magnetite (Fe_3O_4). Iron is abundant in the Earth's crust. It can be extracted relatively easily from its ores and has useful physical properties; therefore iron has become a metal of considerable importance.

- 1 Use figure 1 to give the atomic mass (mass number) and atomic number of iron.

Iron is a typical transition metal.

- It has more than one stable oxidation state in its compounds (+2 and +3)
- It can act as a catalyst (used in the Haber process – ammonia manufacture)
- It forms a variety of complex ions.

56
26 Fe

Figure 1

- 2 Where are the transition metals located in the Periodic Table?

Iron is produced in a blast furnace. This is a tapered, cylindrical tower approximately 30–40 metres high. It is constructed of steel and is lined with special (refractory) bricks that can cope with the high temperatures involved. A blast furnace works continuously and can produce more than 1000 tonnes of iron per 24 hours. The raw materials are iron ore (Fe_2O_3), coke (C, carbon), limestone (CaCO_3 , calcium carbonate) and air (O, oxygen). In the first stage the iron ore is crushed and purified. The iron ore can be anywhere from 30% to 95% iron (III) oxide. The impurities are mainly silicon (IV) oxide (SiO_2 , silica) and aluminium oxide (Al_2O_3 , alumina). A mixture of iron ore, coke and limestone is added at the top of the blast furnace. Hot air is blasted through 'tuyères' (pipes) near the base of the furnace. The air used to supply the blast in a blast furnace is preheated to temperatures approximately 540°C to 870°C . Oxygen from the air reacts with coke exothermically (heat given out) to form carbon dioxide (CO_2).

- 3 Write a balanced equation for the reaction between carbon and oxygen forming carbon dioxide.

The waste gases from the furnace are used to heat the air. The mass of air used in the operation of a blast furnace is more than the total mass of the other raw materials used. The combustion of coke raises the temperature at the bottom of the furnace to approximately 1700°C . As the carbon dioxide rises up the furnace it reacts with more coke to form carbon monoxide (CO) in an endothermic (heat taken in from environment) reaction.

- 4 Write a balanced equation for the reaction between carbon dioxide and carbon forming carbon monoxide.

The carbon monoxide then reduces the iron oxides to iron (removes oxygen) in the upper part of the furnace where the temperature ranges between 300°C and 700°C .

- 5 Write a balanced equation for the reaction between iron oxide and carbon monoxide.

The molten iron falls to the base of the furnace. It is 'tapped off' as a liquid every few hours. The role of limestone is to extract impurities from the iron. When it is heated inside the furnace, limestone decomposes into calcium oxide (CaO) and carbon dioxide.

- 6 Write a balanced equation for the thermal decomposition of limestone.

The calcium oxide then reacts with impurities such as silica and alumina. It forms calcium silicate (CaSiO_3) and calcium aluminate (CaAl_2O_4) respectively.

- 7 Write a balanced equation for the reaction between calcium oxide and silica.
- 8 Write a balanced equation for the reaction between calcium oxide and alumina.

The liquid mixture of silicates flows to the bottom of the furnace and floats on the molten metal. It is tapped off separately from the iron. This **slag** is used in making cement, roads and building materials. The iron produced is still relatively impure and is known as **pig iron**.

- 9 Explain what is meant by each term.

a Native b Abundant c Refractory d Endothermic

- 10 Give three typical properties of a transition metal.

- 11 Give two uses of *slag*.

- 12 What is *pig iron*?

Sodium carbonate (Na_2CO_3 , 'washing soda') is a very important industrial chemical. It is a white powder and is soluble in water. **Hard water** is water that contains a lot of magnesium or calcium salts (and therefore doesn't lather soap well). Sodium carbonate is used to soften hard water and in making soap and glass. The Solvay (ammonia-soda) process was invented by Ernest Solvay (1838–1922) in the mid 1860s. It is a multistage process that involves the recycling of most of the by-products involved in the manufacture of the sodium carbonate.

The raw materials for the Solvay process are limestone (calcium carbonate, CaCO_3) and sodium chloride (NaCl). Ammonia (NH_3) is also used but this is recycled; only small amounts are added at intervals to replenish that lost during the process. It is a cyclical procedure, but the following sequence will help in understanding the overall process.

A Limestone is heated (often by burning coke, a form of carbon) forming calcium oxide (lime, CaO) and carbon dioxide (CO_2).

1 Write a balanced equation for reaction A.

B Ammonia and carbon dioxide are passed into a saturated sodium chloride solution forming ammonium chloride (NH_4Cl) and a precipitate of sodium hydrogen carbonate (sodium bicarbonate, NaHCO_3).

2 Write a balanced equation for reaction B. (Hint: include water as a reactant).

C The temperature is kept below 15°C to ensure that the sodium hydrogencarbonate (slightly-soluble) doesn't dissolve and can be filtered off.

D The sodium hydrogencarbonate is heated to form sodium carbonate, carbon dioxide and water.

3 Write a balanced equation for reaction D.

E The carbon dioxide produced is recycled.

F The calcium oxide is added to water forming calcium hydroxide (slaked lime, $\text{Ca}(\text{OH})_2$) solution.

4 Write a balanced equation for reaction F.

G The ammonium chloride solution is treated with calcium hydroxide solution, forming ammonia, water and calcium chloride (CaCl_2).

5 Write a balanced equation for reaction G.

H The ammonia is recycled.

I Calcium chloride is a by-product.

J The overall reaction is sodium chloride solution and calcium carbonate solid react to form sodium carbonate solid and calcium chloride solution.

6 Write a balanced equation for reaction J.

Figure 1 shows a simplified version of the major reactions in the Solvay process.

7 Copy the diagram but replace each name with the relevant chemical formula.

8 Give the names of the raw materials used as inputs for the Solvay process.

The Solvay process is not without its drawbacks. The process generates large quantities of heat that have to be dealt with. Ammonia is an air pollutant and must not be allowed to escape. There is not a great demand for calcium chloride.

9 Give three uses of sodium carbonate.

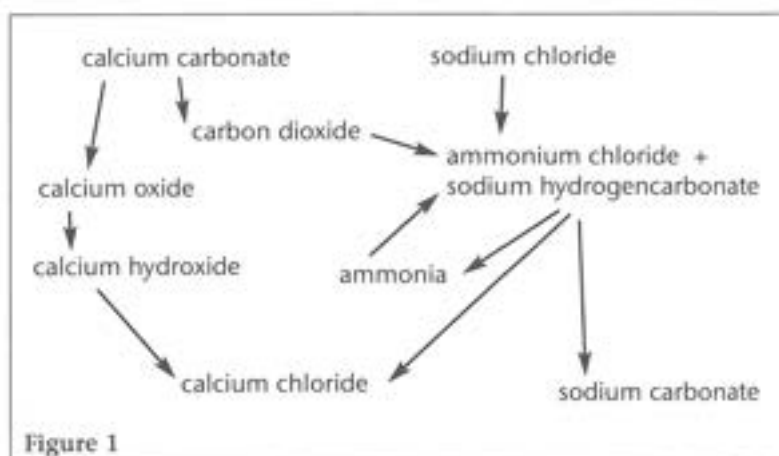


Figure 1

Redox is an abbreviation for the terms **reduction** and **oxidation**. Like many expressions in chemistry, their meanings of reduction and oxidation have evolved. Because oxygen is involved in so many common reactions (such as burning, corrosion, respiration and photosynthesis) many chemists have studied it. Oxidation originally meant the addition of oxygen to an element; reduction originally meant the removal of oxygen. Gradually, electronic theory and the concept of valency expanded, and oxidation and reduction have been related to the transfer or rearrangement of electrons. This in turn has led to the ideas of reduction and oxidation being applied to most reactions.

- 1 The property of elements that determines the ratio of combining atoms of different elements. What does this describe?

Here are some broader definitions of **oxidation**:

- The complete or partial removal of electrons from an atom
- The loss of electrons.

Here are some broader definitions of **reduction**:

- The complete or partial addition of electrons to an atom
- The gain of electrons.

There seems to be a contradiction here. Electrons are particles with a single charge – how can they be **partially** removed? The answer lies in the following descriptions:

- **Ionic bonding** involves one atom transferring one or more electrons to a different atom
- **Covalent bonding** has atoms sharing electrons
- **Polar covalent bonding** has unequal sharing of electrons.

Polar covalent bonding shows characteristics of both ionic and covalent bonding. Electrons are shared, but one atom has a greater share of the electron pair.

- 2 Define each type of bonding

a Ionic b Covalent c Polar covalent

- 3 Discuss this statement: *The processes of reduction and oxidation must take place simultaneously.*

Electronegativity This is a concept that helps in understanding redox reactions. Electronegativity is a measure of how strongly an atom attracts electrons. When two or more atoms are 'competing' for electrons, the more electronegative atoms will 'win', gain electrons and be reduced. The most electronegative elements are Group VII (the halogens) and the least electronegative elements are the Group I elements (the alkali metals).

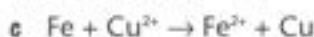
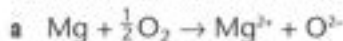
- 4 What is the name of the process where an atom loses electrons?

- 5 Write the term that matches each description.

- a An atom or molecule that readily *captures* electrons.
- b An atom or molecule that readily *releases* electrons.

Reactive metals are oxidised more easily than unreactive metals. This is the reaction between sodium and chlorine: $\text{Na} + \text{Cl}_2 \rightarrow \text{Na}^+ + \text{Cl}^-$. The sodium atom has lost an electron and formed a positive ion. It has been **oxidised**. Chlorine has gained an electron and formed a negative ion; it has been **reduced**.

- 6 For each reaction, identify which atoms have been oxidised and which have been reduced.



- 7 Which Group of elements contain the most electronegative elements?

- 8 Which Group of elements contain the least electronegative elements?

- 9 Name three types of reactions that involve oxygen.

Oxidation number (state)

The oxidation number of an atom is related to its **valency**. (Valency tells the number of electrons involved in the bonding of a particular atom.) When we work out oxidation numbers, we assume that all bonds involve the complete transfer of electrons. The oxidation number tells whether an atom has gained or lost control of electrons and also the number of electrons involved. Because electrons have a negative charge, an atom that loses electrons will have a positive oxidation number; an atom that gains electrons will have a negative oxidation number. Note that the terms **oxidation number** and **oxidation state** are usually regarded as referring to the same thing.

The following rules apply in regard to oxidation numbers:

- All bonding between different types of atoms is regarded as **ionic bonding**
- Atoms in the elementary state (i.e. not bonded with other elements) are assigned an oxidation number of zero
- A positive sign (+) or a negative sign (-) is put before the oxidation number
- A positive oxidation number means an atom has lost electrons
- A negative oxidation number means an atom has gained electrons
- The oxidation number of an ion is the same as its ionic charge number
- Hydrogen has an oxidation state of +1 when bonded with non-metals
- Oxygen usually has an oxidation number of -2 except in peroxides
- Group I metals have an oxidation number of +1
- Group II metals have an oxidation number of +2
- Group VII elements usually have an oxidation state of -1 when bonded with metals
- Transition metals have variable oxidation states
- The algebraic sum of the oxidation numbers of atoms in a compound is zero
- If an atom has more than one possible oxidation number, the actual number is shown by Roman numerals in brackets immediately after the name of the atom in a formula. For example, in Cu (II), copper is in the +2 oxidation state.

We can use these rules to calculate the oxidation number of a particular element in a compound.

EXAMPLES

i What is the oxidation state of oxygen in hydrogen peroxide?

Hydrogen peroxide has the formula H_2O_2 .

The overall charge is zero, so the oxidation numbers for each element (hydrogen and oxygen) must add up to zero.

The oxidation number of hydrogen is +1.

There are two hydrogen atoms, so the sum of the oxidation number for hydrogen is $(2 \times +1) = +2$.

This means that oxidation numbers for oxygen must equal -2. There are two oxygen atoms, so each individual oxygen atom must have an oxidation number of $(-2 \div 2) = -1$.

ii What is the oxidation state of manganese in potassium permanganate?

Potassium permanganate has the formula KMnO_4 .

The overall charge is zero, so the oxidation numbers of potassium, manganese and oxygen atoms must add up to zero.

The oxidation number of potassium is +1. (Potassium is a Group I metal.) There is one potassium atom, so the oxidation number for potassium atoms is: $1 \times +1 = +1$.

Manganese is a transition metal, so the oxidation number is not immediately apparent.

The oxidation number of oxygen is -2. There are four oxygen atoms, so the oxidation number for oxygen atoms is: $4 \times -2 = -8$.

Add up the oxidation numbers: $\text{K} = +1$, $\text{Mn} = ?$, $\text{O}_4 = -8$. $+1 + ? - 8 = 0$; hence $\text{Mn} = +7$.

10 Calculate the oxidation number of each given element. Show your working.

- | | |
|---|---|
| a Nitrogen in ammonia (NH_3) | b Nitrogen in nitric acid (HNO_3) |
| c Sulfur in hydrogen sulfide (H_2S) | d Sulfur in sulfuric acid (H_2SO_4) |

Transition metals can exhibit more than one oxidation state. To show the oxidation state of the transition metal atom in a compound, roman numerals in brackets are used. For example, copper (II) sulfate means that copper is present in the +2 oxidation state in this compound.

- 1 Copper (II) sulfate has the formula CuSO_4 . Work out the oxidation state of sulfur in this compound. Show your working.
- 2 Give the name of each oxide and work out the oxidation state of chromium.
 - a CrO
 - b Cr_2O_3
 - c CrO_3

The oxidation state can help us work out the formula of some compounds of transition metals.

EXAMPLE

- i What is the formula of iron (II) chloride?

The (II) means that iron is in the +2 oxidation state. Hence the oxidation state of the chlorine atoms present must total -2 . Chlorine atoms have the oxidation state of -1 when combined with metals; hence there must be two chlorine atoms present. The formula of iron (II) chloride is FeCl_2 .

- 3 Give the formula for each compound.
 - a iron (III) oxide
 - b cobalt (II) hydroxide

Change in oxidation number Definitions of oxidation and reduction can refer to oxidation number. Oxidation is an increase in oxidation number. Reduction is a decrease in oxidation number.

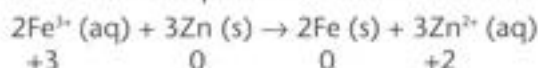
EXAMPLES

- ii Sulfur dioxide and oxygen react to form sulfur trioxide: $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightarrow 2\text{SO}_3(\text{g})$. Has sulfur been oxidised or reduced?

Study the oxidation states of sulfur in the oxides: In SO_2 sulfur has an oxidation state of $+4$ (oxygen is 2×-2); in SO_3 sulfur has an oxidation state of $+6$ (oxygen is 3×-2). The oxidation number of sulfur has changed from $+4$ to $+6$. This is an increase, so sulfur has been oxidised.

- iii When zinc metal is added to a solution containing Fe^{3+} ions, solid iron and aqueous Zn^{2+} ions are formed. Which elements have been oxidised and which have been reduced?

Write the balanced equation with the oxidation numbers underneath:



Iron's oxidation number has decreased from $+3$ to 0 ; therefore iron has been reduced. Zinc's oxidation number has increased from 0 to $+2$; therefore zinc has been oxidised. Iron is the oxidising agent (gained electrons) and zinc is the reducing agent (lost electrons).

- 4 Work out the oxidation numbers of atoms in each reaction then identify which atoms have been oxidised and which have been reduced.
 - a $2\text{Na}(\text{s}) + \text{CuCl}_2(\text{s}) \rightarrow \text{Cu}(\text{s}) + 2\text{NaCl}(\text{s})$
 - b $\text{Zn}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{ZnCl}_2(\text{aq}) + \text{H}_2(\text{g})$
 - c $\text{SiO}_2(\text{s}) + 2\text{Mg}(\text{s}) \rightarrow \text{Si}(\text{s}) + 2\text{MgO}(\text{s})$
 - d $\text{H}_2\text{S}(\text{g}) + \text{CuCl}_2(\text{aq}) \rightarrow \text{CuS}(\text{s}) + 2\text{HCl}(\text{aq})$
 - e $2\text{KClO}_3(\text{s}) \rightarrow 2\text{KCl}(\text{s}) + 3\text{O}_2$ (Hint: in KClO_3 , chlorine is not bonded to a metal, so the oxidation number is not -1)

We can also use oxidation numbers to help balance reactions.

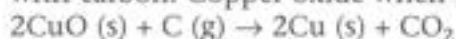
EXAMPLE

- iv When iron filings are added to a solution of copper (II) sulfate, copper metal is deposited and iron (III) sulfate solution is formed. Write a balanced equation for this reaction.

Copper (II) is Cu^{2+} . The sulfate ion has the formula SO_4^{2-} , hence copper (II) sulfate has the formula CuSO_4 . Iron (III) is Fe^{3+} . Hence iron (III) sulfate has the formula $\text{Fe}_2(\text{SO}_4)_3$, ($2 \times +3 = 3 \times -2$). The equation is: $2\text{Fe}(\text{s}) + 3\text{CuSO}_4(\text{aq}) \rightarrow 3\text{Cu}(\text{s}) + \text{Fe}_2(\text{SO}_4)_3(\text{aq})$.

- 5 Write a balanced equation for each of these redox reactions.
 - a When copper wire is placed in a solution of silver nitrate (AgNO_3), silver metal and copper (II) nitrate are formed.
 - b Titanium (IV) chloride forms titanium metal and sodium chloride when heated with sodium.
 - c Lead metal forms lead (II) chloride and hydrogen when reacted with concentrated hydrochloric acid.

Reduction and metal ores Most metals occur combined with non-metals as compounds, such as carbonates, oxides or sulfides. In order to extract a metal from its ore, the metal has to be reduced. Some of the less reactive metals (e.g. copper, tin and lead) can be obtained from their oxides by heating them with carbon. Copper oxide when heated with carbon gives copper metal and carbon dioxide:



- 6 Work out the oxidation numbers for the atoms in the above reaction. Explain why it is clear that the copper has been reduced.
- 7 When tin (IV) oxide is heated with carbon, tin and carbon dioxide are formed.
 - a Write a balanced equation for the tin (IV) oxide reaction.
 - b Show the oxidation states of the atoms in your balanced equation.

Not all metals can be reduced with carbon. Some need a stronger reducing agent such as carbon monoxide (CO). Carbon monoxide can be made by heating a mixture of carbon dioxide and carbon.

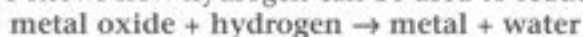
- 8 Write a balanced equation for the synthesis of carbon monoxide.
- 9 Carbon monoxide reacts with iron (III) oxide to form iron and carbon dioxide.
 - a Write a balanced equation for the carbon monoxide reaction.
 - b Show oxidation states of the atoms in your balanced equation.

Although carbon and carbon monoxide are the most common industrial reducing agents, other substances can be used in the laboratory. These include aluminium, hydrogen and methane.

Powdered aluminium can also be used to reduce iron (III) oxide in the **thermit** reaction. Thermit (also spelled thermite) describes a powdered mixture used in welding. Considerable heat is produced by this reaction.

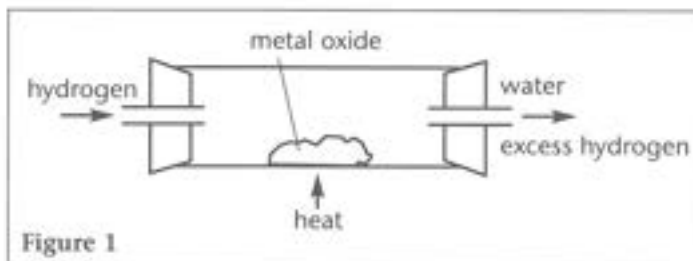
- 10 Write a balanced equation for the thermit reaction and show the oxidation states of the atoms.

Figure 1 shows how hydrogen can be used to reduce a metal oxide. The general equation is:



Hydrogen will only reduce metals found below it in the **Electrochemical (Reactivity) series**.

- 11 Write a balanced equation for the reduction of copper (II) oxide by hydrogen.
- 12 Methane (CH₄) can reduce lead (IV) oxide to lead, carbon dioxide and water. Write a balanced equation for this reaction.



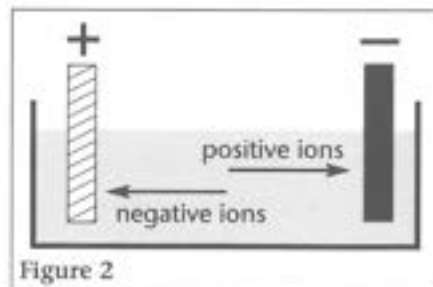
Electrolysis Some metals can only be commercially obtained from their ores by the use of electrolysis. This is the chemical breakdown of compounds by an electric current. For electrolysis, the compound must be in liquid form, either dissolved in water or molten. An electrolytic cell (see figure 2) has two electrodes placed in the compound. When the electric current is switched on, mobile ions are attracted to the electrode of opposite electrical charge. (Remember, 'opposites attract'.)

Aluminium oxide (Al₂O₃) is a solid with a high melting point and has Al³⁺ and O²⁻ ions present. When it is heated and forms a liquid these ions become mobile. In an electrolytic cell, Al³⁺ ions are attracted to the negative electrode and O²⁻ ions are attracted to the positive electrode. At the negative electrode, Al³⁺ ions gain electrons and form Al atoms. At the positive electrode, O²⁻ ions give up electrons and form oxygen atoms.

- 13 Write a balanced equation for the reaction at the negative electrode where aluminium ions gain electrons and are reduced.
- 14 Write a balanced equation for the reaction at the positive electrode where oxygen ions lose electrons and are oxidised.

Your answers to questions 13 and 14 are called **half equations** (or half reactions). They can be added (and balanced) to give the complete reaction.

- 15 Write a balanced equation for the electrolysis of aluminium oxide forming aluminium and oxygen.



Oxidising agents (oxidants) are substances that oxidise (take electrons away from) other substances. Common oxidising agents include potassium dichromate, concentrated tetraoxosulfuric acid, concentrated nitric acid and potassium permanganate.

- Potassium dichromate has the formula $K_2Cr_2O_7$. Work out the oxidation state of chromium in this compound.
- Tetraoxosulfuric acid (more commonly known as sulfuric acid) has the formula H_2SO_4 . Work out the oxidation state of sulfur in this compound.
- Potassium permanganate has the formula $KMnO_4$. Work out the oxidation state of manganese in this compound.

Potassium dichromate (VI) in an acidic solution oxidises alkanols (alcohols) to the corresponding alkanolic (carboxylic) acids; e.g. ethanol is oxidised to ethanoic (acetic) acid. (See figure 1.)

- Give the molecular formula for ethanol.
- Give the molecular formula for ethanoic acid
- In this reaction, the orange dichromate (VI) ions are changed to green chromium (III) ions. Is chromium oxidised or reduced? Explain your answer.

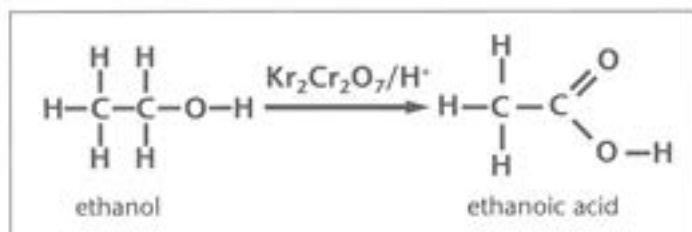
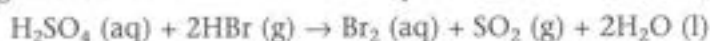


Figure 1

Hydrogen bromide can be oxidised by a concentrated solution of sulfuric acid:



- Copy the equation above and calculate the oxidation states of the atoms present.

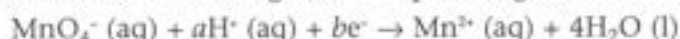
If copper metal is added to concentrated nitric acid, then water, copper nitrate and nitrogen dioxide can be formed:



- Copy the equation above and calculate the oxidation states of the atoms present.

Potassium permanganate is a strong oxidising agent. Solutions of potassium permanganate are useful for measuring concentrations of reducing agents. When potassium permanganate is dissolved in water it forms aqueous potassium (K^+) and manganate (MnO_4^-) ions. Potassium permanganate is frequently used in acidified solution and is purple in colour. When it is reduced, it forms the colourless manganese (Mn^{2+}) ion.

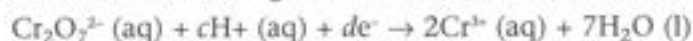
A typical reaction involving acidified permanganate solution is shown in part below:



- Calculate the number of hydrogen ions (a) present in the left side of the equation.
- Calculate the number of electrons (b) present in the left side of the equation.

Acidified dichromate ($Cr_2O_7^{2-}$) ions behave in a similar manner to permanganate ions. Dichromate ions are reduced to chromium (Cr^{3+}) ions.

A typical reaction involving acidified dichromate solution is shown in part below:



- Calculate the number of hydrogen ions (c) present in the left side of the equation.
- Calculate the number of electrons (d) present in the left side of the equation
- Write a balanced equation for each of these redox reactions. Show the oxidation states of the atoms present.
 - One process that occurs in a lead-acid accumulator (car battery) is lead metal and lead (IV) oxide in sulfuric acid producing lead (II) sulfate and water.
 - When potassium dichromate solution is added to hydrochloric acid, chlorine gas, water, chromium (III) chloride and potassium chloride are formed.

Galvanic cells Redox reactions involve the transfer of electrons, so these reactions can be used to generate an electric current. Electrochemical cells that generate electricity are called Galvanic cells, named in honour of Luigi Galvani (1737–1798). Galvani investigated the effect of electricity on muscles. In turn, his research led Alessandro Volta (1745–1827) to develop (around 1800) the first continuous chemical source of electricity. Volta placed discs of silver, cardboard soaked in brine (sodium chloride solution) and zinc in repeating layers. One end had a silver plate on the outside, the other end had zinc. When he joined the silver and the zinc with a wire, electricity flowed continuously through the wire. Figure 2 shows one way of looking at this experiment in a school laboratory. If separate zinc and silver electrodes are placed in an electrolyte such as sodium chloride solution, reactions occur and a current flows. (An **electrolyte** is a liquid that allows a current to flow because there are mobile ions present.) The wire connecting the electrodes (via a voltmeter) allows electrons to flow. The following observations can also be made:

- The zinc electrode is negative
- The silver electrode is positive
- The electrodes change in appearance.

14 What evidence is there to show that electrons flow from the zinc electrode to the silver electrode?

15 Redraw figure 2, showing the charges on both electrodes and the direction of current from zinc electrode to silver electrode in the external circuit.

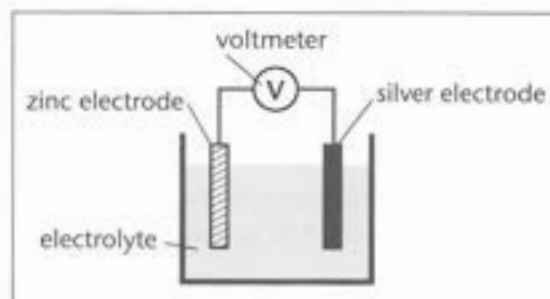


Figure 2

The simple cell shown in figure 2 generates current for only a short time because an imbalance of charge builds up. In order to get a clearer idea of the reactions involved, a more complex cell has to be set up. Figure 3 shows a zinc electrode in a solution of a zinc salt and a silver electrode in a solution of a silver salt. A salt bridge (e.g. KNO_3 or NH_4Cl , gel or solution) allows ions to move (and hence completes the circuit) but minimises the contamination of the two solutions. Ions need to move to balance the flow of electrons.

The following observations can also be made:

- The zinc electrode loses mass
- The silver electrode gains mass.

16 Redraw figure 3, showing the charges on both electrodes and the direction of current from zinc electrode to silver electrode.

17 Outline the purpose of the salt bridge.

18 Name a suitable substance for use as a salt bridge.

19 At the zinc electrode, zinc metal is forming zinc (Zn^{2+}) ions.

Write an equation for this reaction. State whether it is oxidation or reduction.

20 At the silver electrode, silver (Ag^+) ions are forming silver metal. Write an equation for this reaction. State whether it is oxidation or reduction.

21 The answers to Q19 and Q20 are *half-equations*. Adjust your equations for Q19 and Q20 so that they have the same number of electrons present. Add the two half-equations to show the overall reaction.

22 Reduction takes place at the *cathode* and oxidation takes place at the *anode*.

- Which electrode is the cathode?
- Which electrode is the anode?

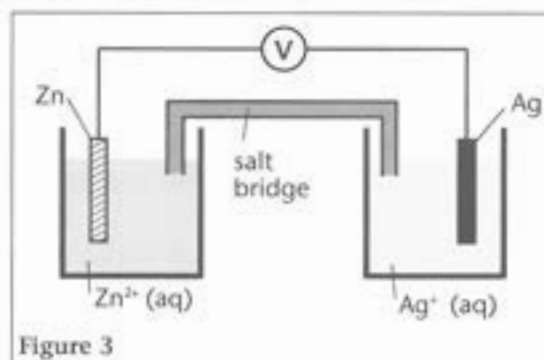


Figure 3

A **galvanic cell** consists of two **half-cells**. These half-cells each consist of an electrode and an electrolyte. The electrolyte usually contains ions in aqueous solution of the element making up the electrode. One half-cell is a reduction cell and the other is an oxidation cell. When a metal is placed in a solution of ions of that metal, it tends to form more ions (as shown in figure 1).

This can be shown as an equation:



For example, if zinc metal is placed in a solution of zinc (II) sulfate, then the zinc metal forms positive ions.

- 1 Write an equation to show the formation of zinc ions in aqueous solution.
- 2 If silver metal is placed in an aqueous solution of silver nitrate, the silver forms silver ions. Write an equation to show the formation of silver ions in aqueous solution.

If the two half-cells described above are connected to form a cell (as shown in figure 2), there seems to be a contradiction. Both metals 'want' to form positive ions and give off electrons. This means that there will be a 'competition', as electrons can only flow one way through the circuit at a time. The more reactive (**electropositive**) metal 'wins', forming ions and giving out electrons. The lesser reactive metal 'loses' and its ions gain electrons and form the metal. In this case, zinc is a more reactive metal than silver is. This means that electrons flow from the zinc electrode through the external circuit, via the voltmeter, to the silver electrode. Silver ions in solution gain electrons and form silver metal. The zinc electrode is negative and the silver electrode is positive.

- 3 Redraw figure 2, showing the direction of electron flow and the electrical charge on the electrodes.
- 4 As the reaction progresses, what will happen to the mass of the:
 - a zinc electrode?
 - b silver electrode?

This cell can also be represented as:



The two vertical lines (||) represent a salt bridge. The single vertical line (|) represents the boundary between the metal electrode and electrolyte. We can read the equation from left to right: zinc metals form (dipositive) zinc ions and (unipositive) silver ions form silver metal. In this convention, the left hand electrode (zinc) is negative and the right hand electrode (silver) is positive.

- 5 Write each cell equation in words.
 - a $Mg | Mg^{2+} || Fe^{2+} | Fe$
 - b $Ni | Ni^{2+} || Cu^{2+} | Cu$
- 6 Copy the passage and use these words to complete it: *charge, connected, electrode, electrons, electropositive, galvanic, oxidation, positive, salt*.

A (a) cell generates electricity by means of chemical reactions. Two half-cells are required; in one half-cell reduction takes place and in the other half-cell (b) occurs. A half-cell consists of a metal (c) dipping into an electrolyte. The electrodes of the half-cells are (d) by an external circuit through which (e) flow. The electrolytes are connected by a (f) bridge through which ions move to avoid building up electrical (g). Metals tend to lose electrons and form (h) ions, but this can only happen for one of the electrodes. The more reactive (i) metal is the negative electrode.

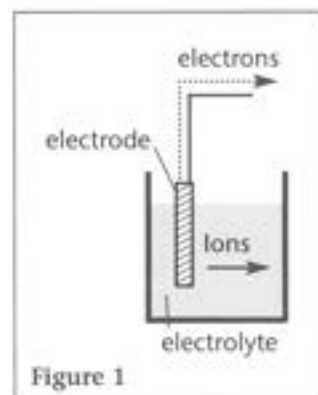


Figure 1

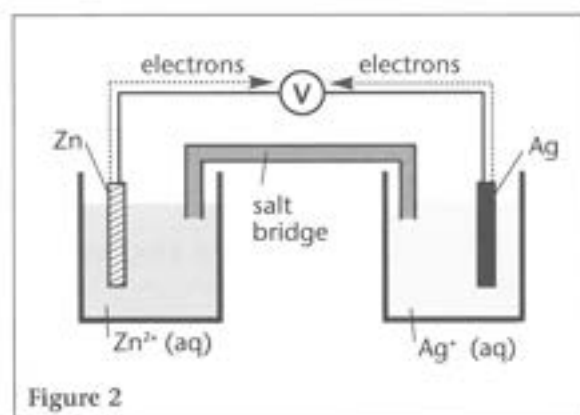
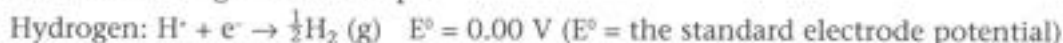


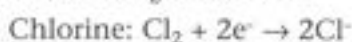
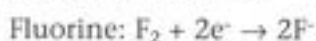
Figure 2

Standard electrode potentials The reactions (and voltages) produced by a pair of different electrodes can be predicted using a table of standard electrode (reduction) potentials. This table compares how ready a species is to accept electrons (be reduced) compared to a standard hydrogen electrode. (Species is the general term for positive and negative ions.) The redox couple $\text{H}^+(\text{aq}) \rightarrow \frac{1}{2} \text{H}_2(\text{g})$ is assigned a voltage of zero under specific conditions.

Formats vary, but the convention is to show the oxidised form of the redox couple on the left and the reduced form on the right. For example:



Study the following equations for redox couples:



Note that the spontaneous direction of the reaction is not necessarily that shown in the redox couple. As a rough guide to predicting reactions, a negative E° means that the spontaneous direction of the reaction (compared to hydrogen) is right to left.

7 Write a similar equation for each element.

- Bromine
- Aluminium
- Copper (assume it has +2 oxidation state)
- Iodine

The original standard electrode used to measure standard electrode potentials is cumbersome. A version of the standard hydrogen electrode (SHE) is shown in figure 3. The concentration of hydrogen ions is 1 mole per litre, the gas pressure is 100 kPa (or sometimes 101.3 kPa, 1 atmosphere) and the temperature is 25°C (298K).

The **standard hydrogen electrode** is an electrode in which hydrogen gas bubbles over a platinum surface in contact with aqueous hydrogen ions under specific conditions.

Other reference electrodes tend to be used because they are easier to operate.

8 Copy figure 3 and label:

- hydrogen in
- hydrogen out
- hydrogen ions
- platinum electrode
- voltmeter.

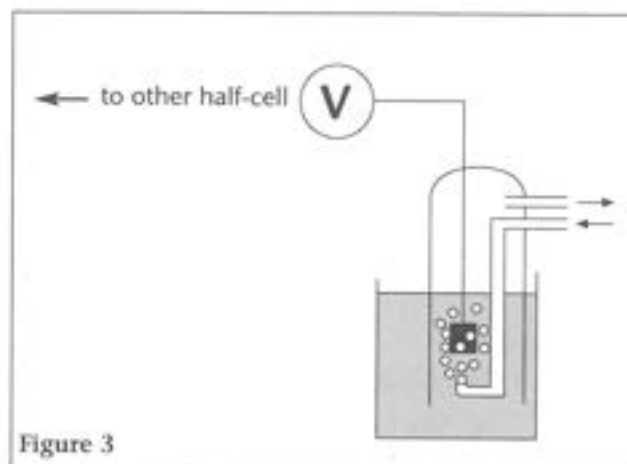


Figure 3

9 A table of reduction potentials shows sodium as having this half-reaction: $\text{Na}^+ + \text{e}^- \rightarrow \text{Na}$ $E^\circ = -2.7 \text{ V}$ Is this reaction (when linked to a hydrogen half-cell) likely to proceed from left to right as shown? Explain your answer.

10 Match each description to the correct term: *anode, cathode, electrolyte, salt bridge, voltmeter*.

- Measures the electromotive force (EMF) in volts generated by two half-cells.
- The electrode at which oxidation occurs.
- The electrode at which reduction occurs.
- Completes the electrical circuit in a Galvanic cell.
- A solution containing mobile ions.

Acids

The definitions of acids and bases have changed over the years. One way of classifying substances was by taste: sweet, sour, salt or bitter. After a while it was noticed that the sour substances had many other properties in common; e.g. they changed the colour of blue litmus to red and corroded some metals. The Latin verb *acere* means 'to make sour'. This gave the term **acid**. Some common, naturally-occurring acids that have been known for a long time include ethanoic acid (acetic acid or vinegar, CH_3COOH), citric acid (from citrus fruits, 2-hydroxypropane-1, 2, 3-tricarboxylic acid, see figure 1) and lactic acid (from sour milk, $\text{CH}_3\text{CH}(\text{OH})\text{COOH}$).

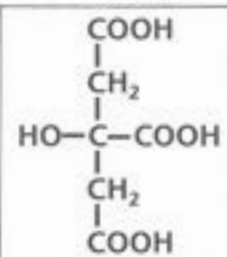


Figure 1

- 1 What is the general term for substances such as litmus that change colour in the presence of acids and bases? Why do you think they are called this?
- 2 Give the molecular formula for citric acid. (See figure 1.)

In 1661, Robert Boyle (1627–1691) published a work called *The Sceptical Chymist* in which, among other things, he distinguished between acids and bases (although he referred to bases as alkalis). He noted that all acids turned a particular vegetable indicator from blue to red and all alkalis turned the indicator green. Boyle found that some substances did not change the colour of this indicator and he concluded that these substances were neutral. Around this time it was also proposed that salts are formed when acids react with bases.

Antoine Laurent Lavoisier (1743–1794) was one of the first people to investigate what was responsible for giving an acid its properties. By 1776 he had come to the conclusion that one element was present in all of the acids he knew; hence this element must cause an acid to behave as it does. This element was **oxygen**, named from the Greek terms *oxein* (sour) and *gennen* (to generate). Lavoisier was wrong, because he only knew the formulae of a few acids.

Around 1815, Sir Humphry Davy (1778–1829) came to the conclusion that acidity was not caused by oxygen because he had discovered that there were acids that did not contain any oxygen. Acidity must be caused by something else.

- 3 Give the names and sources of three naturally-occurring acids.

Davy showed that muriatic acid (now called hydrochloric acid) could be reacted with mercury to give only mercury chloride and hydrogen as products, no oxygen.

- 4 Balance this equation that shows Davy's experiment: $\text{HCl} + \text{Hg} \rightarrow \text{Hg}_2\text{Cl}_2 + \text{H}_2$

Davy suggested that acidity was caused by hydrogen, because this element was common to all acids.

- 5 If all acids contain hydrogen, does this mean that all compounds containing hydrogen are acids? Explain your answer.

Around 1840, Baron Justus von Liebig (1803–1873) came up with the first adequate definition of an acid: *A compound containing hydrogen in a form in which it can be replaced by a metal*. This classification is still very useful today.

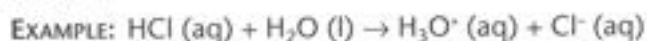
Bases

Bases were thought of as simply compounds that could neutralise acids. This was until Svante August Arrhenius (1859–1927), came up with a more complex theory in 1887. He concluded that acids are substances that, when dissolved in water, broke up (dissociated) to give hydrogen (H^+) ions. In other words, substances that dissolve in water (dissociate) to produce hydrogen ions are called acids. Bases are substances that produce hydroxide (OH^-) ions when dissolved in water. This is known as the **Arrhenius concept**. Note that the hydrogen ion does not exist by itself in water. It is hydrated and is often written as H_3O^+ , the hydronium ion.

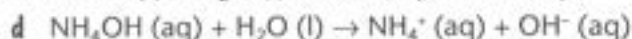
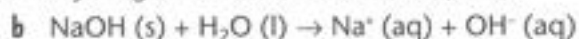
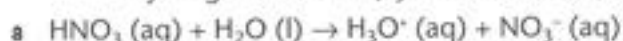
- 6 Robert Boyle classified substances into three groups. What are these groups?
- 7 Which chemist believed that oxygen was the element responsible for a compound being an acid?
- 8 Which chemist is credited with associating hydrogen with acidic properties?
- 9 A substance that can neutralise an acid producing a salt plus water. What does this define?
- 10 Give the formula for the hydrogen ion.
- 11 Give the formula for the hydroxide ion.

Dissociation

12 The following reactions show how some substances behave when they dissolve in water. Copy each equation, name the substances present and state whether the substances are acids or bases according to Arrhenius.



Hydrogen chloride (hydrochloric acid) + water $\rightarrow$ hydrogen ions + chloride ions (acid)



Brønsted–Lowry The Arrhenius concept of acids and bases had some shortcomings:

- Only aqueous solutions were considered
- Only substances containing hydroxides could be bases.

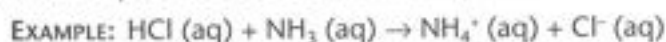
In 1923, two chemists separately and independently came up with broader definitions of acids and bases. Johannes Brønsted (1879–1947) and Thomas Lowry (1874–1936) focused on the role of protons (hydrogen ions). Their definitions:

- Acids are substances from which protons can be removed
- Bases are substances that bind protons.

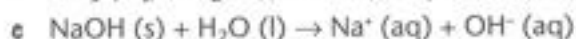
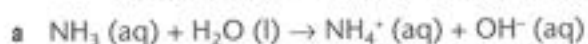
These are often quoted as:

- Acids donate protons
- Bases accept protons.

13 The following reactions show how some acids and bases react. Copy each equation, name the substances present and decide which species are acids and bases according to Brønsted–Lowry.



Hydrochloric acid (HCl) has transferred a proton to ammonia (NH_3) forming ammonium and chloride ions. HCl is the acid and NH_3 is the base.



Note that all Arrhenius acids are acids according to the Brønsted–Lowry definitions, but not all Brønsted–Lowry acids are Arrhenius acids.

14 Copy each statement and decide whether it applies to acids or bases or both.

- | | |
|---|-------------------------------|
| a Have a bitter taste | b Have a sour taste |
| c Aqueous solutions are electrolytes (conduct electricity) | d Have a slippery feel |
| e Turn blue litmus red | f Turn red litmus blue |
| g React with many metals to form hydrogen and a salt | |

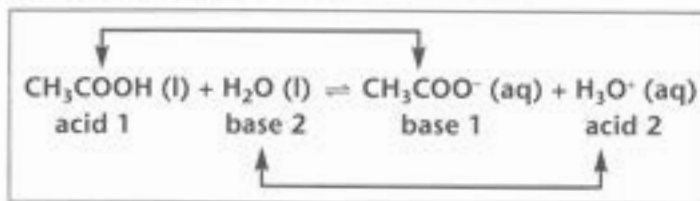
Research 15 An American chemist named Gilbert Newton Lewis (1875–1946) came up with yet another definition of acids and bases. Summarise his definitions of acids and bases.

16 When an acid and a base react, a salt is formed. Sulfuric acid forms sulfates, nitric acid forms nitrates, and so on. Copy and complete the table. The first one has been done for you.

17 Define *salt*.

Acid	Formula	Salts formed
Hydrochloric	HCl	Chlorides
	HBr	
	HNO ₃	
Sulfuric		
		Ethanoates (acetates)

Acid–base conjugate pairs When an acid reacts with a base, a conjugate acid and conjugate base are formed. *Conjugate* means *to join*. Conjugate acid–base pairs only differ by a proton (H^+). Figure 3 shows the reaction of ethanoic acid. Consider the reaction from left to right. Ethanoic (acetic) acid (CH_3COOH) has donated a proton to water (H_2O); hence ethanoic acid is an acid and water is a base. Looking at the reaction from right to left, the hydronium ion (H_3O^+) is an acid because it has donated a proton to the ethanoate (acetate) ion (CH_3COO^-). CH_3COOH and CH_3COO^- are conjugate acid–base pairs. The other conjugate acid–base pair is $\text{H}_3\text{O}^+/\text{H}_2\text{O}$.

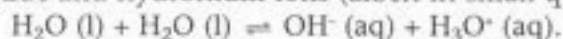


7 Refer to figure 3 to help you work out the conjugate acid–base pair for each reaction.

- a $\text{HNO}_3(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{NO}_3^-(\text{aq})$
- b $\text{HBr}(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{Br}^-(\text{aq})$
- c $\text{H}_2\text{SO}_4(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons 2\text{H}_3\text{O}^+(\text{aq}) + \text{SO}_4^{2-}(\text{aq})$
- d $\text{C}_6\text{H}_5\text{NH}_2(\text{s}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{C}_6\text{H}_5\text{NH}_3^+(\text{aq}) + \text{OH}^-(\text{aq})$
- e $\text{N}_2\text{H}_4(\text{l}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{N}_2\text{H}_5^+(\text{aq}) + \text{OH}^-(\text{aq})$
- f $\text{NH}_3(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{NH}_4^+(\text{aq}) + \text{OH}^-(\text{aq})$

Figure 3

Amphiprotic substances Depending on their chemical environment, some species can donate and also (under different circumstances) accept protons. These are amphiprotic substances. Water forms hydroxide and hydronium ions (albeit in small quantities) as shown by the equation:



One water molecule has donated a proton to another water molecule.

The hydrogencarbonate (bicarbonate) ion has the formula HCO_3^- and only exists in aqueous solution. There is a similar term, **amphoteric**, which refers to substances that can react with either acids or bases.

8 In each reaction, state whether the hydrocarbonate ion is donating a proton or receiving a proton.

- a $\text{HCO}_3^-(\text{aq}) + \text{NH}_4^+(\text{aq}) \rightleftharpoons \text{H}_2\text{CO}_3(\text{aq}) + \text{NH}_3(\text{aq})$
- b $\text{HCO}_3^-(\text{aq}) + \text{H}_2\text{O}(\text{l}) \rightleftharpoons \text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq})$

Polyprotic acids Acids that generate one proton per molecule when they ionise are **monoprotic** acids. Some acids are capable of donating more than one proton per molecule. Sulfuric acid is a **diprotic** acid and phosphoric acid is a **triprotic** acid.

9 Study the equation. Explain what is meant by describing sulfuric acid as a *diprotic* acid.



10 Study the equation. Explain what is meant by describing phosphoric acid as a *triprotic* acid.



11 Mark each statement as true or false.

- a Ions are electrically-neutral particles.
- b Strong acids don't ionise completely when they dissolve in water.
- c Group I metal hydroxides are often strong bases.
- d The common (trivial) name for ethanoic acid is formic acid.
- e In an acid–base conjugate pair, the two species only differ by a proton.
- f Phosphoric acid forms salts called phosphates.
- g All acids only donate one proton per molecule.
- h Acids donate protons, bases accept them.

12 Define each term.

- | | |
|-------------------------------|-------------------|
| a Ionise | b Polyprotic acid |
| c Weak acids and strong acids | d Equilibrium |
| e Amphiprotic substances | |

Concentration of acids and bases

Concentration is a measurement of how much of a solute is present in a solution. (Remember, **solute** is the substance being dissolved.) When dealing with solutions of acids and bases, chemists usually use moles per litre which is written as mol L^{-1} or **M**. Figure 1 shows how to calculate the number of moles of a substance. Figure 2 shows how to calculate the molarity of a solution.

EXAMPLES

- i What is the molarity of a solution of hydrochloric acid when 0.35 moles are dissolved in 500 mL of water?

$$\begin{aligned}\text{Concentration} &= \text{no. of moles} \div \text{volume} \\ &= 0.35 \div 0.5 = 0.70 \text{ mol L}^{-1}\end{aligned}$$

- ii What is the molarity of a solution made by dissolving 2.00 g of sodium hydroxide (NaOH) in 250 mL of water? (Atomic mass of Na = 23.00, O = 16.00, H = 1.00)

$$\text{Molar mass of sodium hydroxide} = (23.00 + 16.00 + 1.00) = 40.00 \text{ g}$$

$$\text{Number of moles of sodium hydroxide} = 2.00 \text{ g} \div 40.00 \text{ g} = 0.05 \text{ moles}$$

$$\text{Molarity} = 0.05 \div 0.25 = 0.2 \text{ mol L}^{-1}$$

$$\text{no. of moles} = \frac{\text{mass present (g)}}{\text{molar mass (g)}}$$

Figure 1

$$\text{concentration (moles per litre)} = \frac{\text{no. of moles}}{\text{volume (litres)}}$$

Figure 2

- 1 Calculate the concentration of each solution in moles per litre.
 - a 2 moles of sulfuric acid in 1.5 litres of solution
 - b 1.75 moles of nitric acid in 600 mL of solution
 - c 12.5 g of hydrochloric acid (HCl) in 250 mL of solution (atomic mass of Cl = 35.5)
 - d 15.5 g of ammonia (NH₃) in 500 mL of solution (atomic mass of N = 14.00)
 - e 2.05 g of barium hydroxide (Ba(OH)₂) in 100 mL of solution (atomic mass of Ba = 137.3)

Concentration and strength

These two terms are often confused. **Concentration** refers to the amount of dissolved substance in a solution. A concentrated solution contains more **solute** than a dilute solution. **Strength** refers to the proportion of dissolved molecules that ionise when they dissolve in water. A strong acid has a higher percentage of molecules ionised in solution than a weak acid does.

Research 2 Give your own definitions of the terms *concentration* and *strength*, in respect to acids and bases. Include diagrams to illustrate your answers.

The pH scale Around 1909, S.P.L. Sørensen introduced the concept of pH while working for the Carlsberg brewery in Copenhagen. This is a way to describe hydrogen ion concentration (and hence acidity) in simple numbers. pH is the negative log (to the base ten) of the hydrogen ion concentration expressed in moles per litre. (See figure 3.) Note that logarithms use powers of ten to transform very large or very small numbers into more manageable units. For example, water is neutral. It has a pH of 7 which is a small, tidy number to work with. If the negative log wasn't used, we'd need to write the pH of water as 1.0×10^{-7} M. A pH of less than seven is **acidic**, a pH of seven is **neutral** and a pH above seven is **basic**.

- 3 Copy and complete the table.

[H ⁺] (moles per litre)	pH	Type of solution
10^{-1}	1	
10^{-4}	4	
10^{-7}	7	
10^{-11}	11	
10^{-14}	14	

$$\text{pH} = -\log_{10} [\text{H}^+]$$

Figure 3

- 4 What relationship do you notice about the hydrogen ion concentration and the pH?

Calculating pH

When carrying out calculations of pH on a scientific calculator, you will need to be familiar with the following operations, $\log_{10}$ (log, *not* ln), exponential (EXP or EE) and sign change (+/-).

Note that, depending on the calculator used, the sequence may differ from below.

EXAMPLES

- iii What is the pH of a solution with hydrogen ion concentration 3.5×10^{-4} moles per litre?

Enter the following:

- LOG
- 3.5
- EXP
- 4
- +/-
- =
- +/- The answer is 3.45.

- iv What is the pH of a solution with hydrogen ion concentration 0.05 moles per litre?

Enter the following:

- LOG
- 0.05
- =
- +/- The answer is 1.30.

- 5 Calculate the pH of each solution. Show your working.

- | | |
|--|---|
| a $[H^+] = 0.40$ moles per litre | b $[H^+] = 0.00001$ moles per litre |
| c $[H^+] = 3.0$ moles per litre | d $[H^+] = 3.2 \times 10^{-3}$ moles per litre |
| e $[H^+] = 9.2 \times 10^{-7}$ moles per litre | f $[H^+] = 3.2 \times 10^{-12}$ moles per litre |

Why do we need to know pH values? pH can help chemists identify chemicals. Many chemical processes need to take place at a specific pH in order to obtain optimum results. Many agricultural crops need soil of a particular pH to give the best yield. Marine life is affected if pH changes due to a discharge of industrial waste. Human skin has a pH of 5.5 and many cosmetics and cleansers have been set to this value to minimise skin irritation. Cell function in the body is dependent on body pH levels being within a certain limit. Enzyme activity is vital for keeping body processes working properly and pH levels directly affect this activity.

- 6 Explain what is meant by *optimum results*.
- 7 What is *marine life*?
- 8 Define *pH*.
- 9 Explain what is meant by a pH of 5.5.

Measuring pH Indicators have long been a way of establishing the approximate pH of a solution. An indicator is a dye that changes colour over a specific pH range. Indicators can be in the form of solutions or strips of paper can be soaked in the indicator. One of the most well-known indicators is **litmus**. This is a dye (known for hundreds of years) that is obtained from lichens. Litmus turns red in acidic solutions and blue in basic solutions. **Universal indicator** is a mixture of several indicators which have an overlap of colour-change over a range of pH.

- Research** 10 Give the names of three different indicators (apart from litmus), their colours and the pH ranges at which they change colour.

- Research** 11 Draw up a table that shows the colours of universal indicator from pH 0 to pH 14.

pH can also be measured electronically. A pH meter consists of a special electrode connected to a meter. It is calibrated by immersing the electrode in solutions of known pH (buffers). pH meters can measure pH more accurately than chemical indicators and can also be connected to data loggers to plot changes in pH against variables such as time. They are more expensive and complex to use than chemical indicators.

- 12 Give two advantages and two disadvantages of pH meters compared to indicators.

- Research** 13 Buffer solutions have uses other than calibrating pH meters. Find out what buffer solutions are and what else they can be used for.

Nuclear chemistry studies the nucleus (centre) of atoms. In 1909, Ernest Rutherford was the first person to come up with the idea that the nucleus of an atom was very small, dense and positively charged. Rutherford calculated that the radius of an atom would be in the order of 10^{-10} m and the radius of the nucleus would be approximately 10^{-14} m.

1 How does the size of the radius of the nucleus compare to that of the whole atom?

Shortly after this (1913) it was calculated that the protons of an individual atom made up about half of the mass of the nucleus. Hence there must be some other (unknown) particles making up the 'missing' mass. The concept of **atomic number** (Z) came into use around this time. The atomic number of a particular atom of an element shows:

- the number of electrons
- the number of protons
- the order in which the element is listed in the Periodic Table.

In 1932, James Chadwick discovered that inside atoms there were uncharged particles with a mass approximately that of a proton. We now call this particle a **neutron** and know that it is located inside the nucleus.

2 Copy and complete the table.

Particle	Mass (amu)	Electrical charge	Position
	5×10^{-4}		In shells outside the nucleus
Neutron		None	
	1.00		In the nucleus

All atoms of an element have the same atomic number (Z), but some atoms have a different atomic mass (A). This is because they can have different numbers of neutrons in the nucleus. Atoms of the same elements that have different masses are called **isotopes**. English physicist Francis Aston (1877–1945) was awarded a Nobel Prize in 1922 for his invention of the **mass spectrograph**. This machine can show the relative percentages present of different isotopes of the same element.

Figure 1 shows a simplified mass spectrometer trace for the element chlorine. Each peak shows the presence of a particular isotope. The height of the peak shows the relative abundance of each isotope.

- 3 How many isotopes of chlorine are shown in figure 1?
- 4 What is the mass of the more common isotope of chlorine?

In nature, chlorine exists as a mixture of chlorine-35 (75%) and chlorine-37 (25%). Figure 2 shows how to calculate the average mass of naturally-occurring chlorine.

5 Use the equation shown in figure 2 to calculate the average relative atomic mass of chlorine. Don't forget to include units.

6 Copper exists in nature as two main isotopes: copper-63 (69%) and copper-65 (31%). Calculate the average relative atomic mass of copper.

7 Match each term to the correct description: *amu*, *isotope*, *mass spectrograph*, *neutron*.

- An atomic particle that has no electrical charge and a mass approximately that of a proton
- A machine that gives the relative atomic masses and abundances of atoms
- An atom of an element that has different numbers of neutrons to other atoms of that element
- The unit of mass used when dealing with atoms and molecules

Did you know? The neutron gets its name from the Latin *neuter* meaning 'neither' because it has neither a negative nor a positive charge.

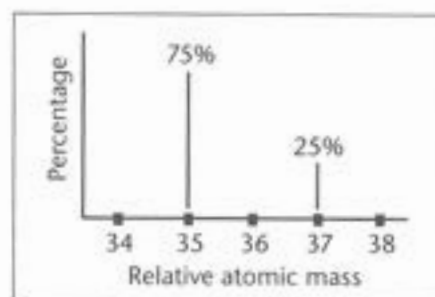


Figure 1

$$\frac{(75 \times 35.0) + (25 \times 37.0)}{100}$$

Figure 2

Radioactivity

Henri Becquerel (1852–1908) studied the properties of uranium compounds. In 1896 he found that they fogged photographic plates even when the plates were wrapped in an opaque covering. The radiation given off by uranium resembled X-rays (discovered by Roentgen in 1895) but could be deflected by electric or magnetic fields, which did not affect X-rays. Hence Becquerel's radiation must be made up of charged particles. Becquerel shared the Nobel Prize for Physics in 1903 with Pierre Curie (1859–1906) and Marie Curie (1867–1934). The term 'radioactivity' was first used by Marie Curie.

Further studies revealed that there were other naturally radioactive elements, such as thorium, radon, radium and polonium. Three main types of ionising radioactivity were identified: alpha (α), beta (β) and gamma (γ). Figure 3 summarises how radiation from radium behaves when it comes into contact with thin sheets of different materials.

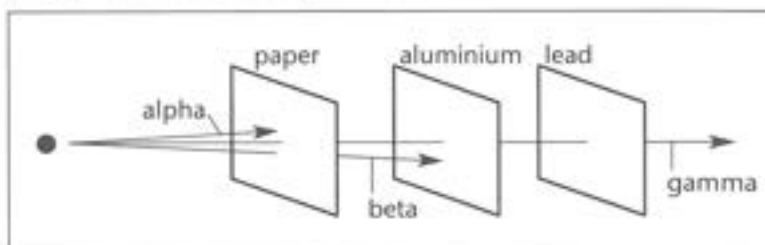


Figure 3

- 8 Give the name and symbol of each radioactive element mentioned so far on this page.

Figure 4 shows what happens when a beam of ionising radiation is placed in a magnetic field. A similar effect is seen if an electric charge is brought close to this radiation. Alpha particles are helium nuclei (${}^4_2\text{He}^{2+}$) and they are made of two neutrons and two protons.

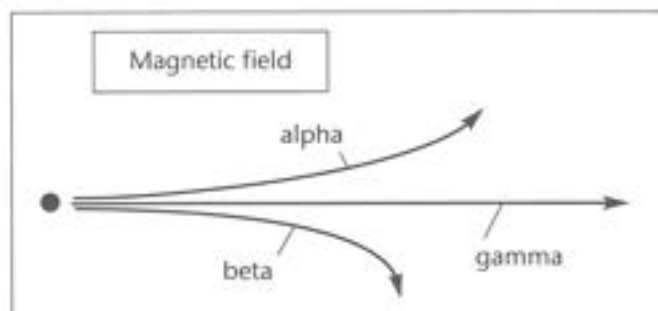


Figure 4

Alpha particles travel at a range of speeds, but around 5% of the speed of light is a typical value. (Light travels at approximately $300\,000\text{ km.s}^{-1}$.) The main source of alpha particles in nature is radon gas, which itself is a breakdown product of uranium.

Beta radiation is a stream of fast-moving electrons (called beta particles) emitted by a nucleus during radioactive decay. These particles have very little mass (about $5.41 \times 10^{-4}\text{ amu}$) and travel close to the speed of light. **Gamma radiation** is at the high frequency (and hence high energy, short wavelength) end of the electromagnetic spectrum. It travels at the speed of light.

Problems with radiation When alpha, beta and gamma radiation collide with neutral particles they can remove electrons from these neutral particles and convert them into charged particles (ions). Exposure to ionising radiation can damage chromosomes and cause cancers and tumours. Alpha particles are relatively heavy, cause a lot of ionisation and can damage body cells. Airborne alpha radiation does not usually pose much of a hazard because it can only travel a short distance in air and is stopped by clothing. If an alpha source is taken into the body (e.g. by contaminated food or water) then it can cause severe problems. Beta particles are much less ionising than alpha particles and are much lighter but are more penetrating. However some radioisotopes that emit beta particles are soluble. Once these are in the food chain they can harm humans if ingested. Gamma radiation, because it has a very short wavelength, will pass through cells usually causing little damage unless the radiation is very intense.

- 9 Draw up a table to summarise the properties of radiation. Use the headings shown below.

Name	Nature of radiation	Speed	Stopped by	Ionising strength

- 10 Explain why alpha, beta and gamma radiation are referred to as *ionising radiation*.
 11 Explain why alpha radiation can cause damage if it enters the human body.

Detecting radiation There are many ways to detect ionising radiation. Two of these ways are with photographic film and with the Geiger-Müller tube.

Workers who could be exposed to radiation often wear **dosimeters** (dosemeters) which are usually a photographic film badge of the type shown in figure 1. This can show how much radiation the wearer has been exposed to over a given period of time (or *doses* of radiation, hence the name). The example shown has a piece of photographic film wrapped in a light-proof cover and placed inside a container that can be clipped to worker's clothing. The film is developed at regular intervals.

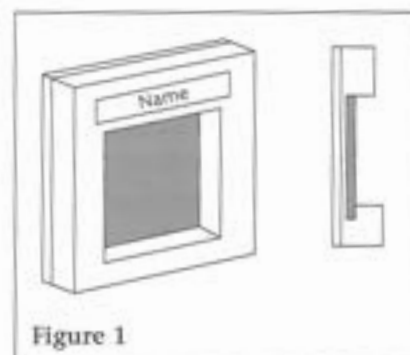


Figure 1

- 1 Copy figure 1 and label the following: *plastic container, light-proof covering, photographic film.*
- 2 Explain how the developed film in the dosimeter indicates the level of radiation exposure of a worker.

Film badges show the cumulative radiation a worker has been exposed to over a period of time, but in many cases we need to know how much radiation is currently being emitted. A **Geiger-Müller tube and counter** (usually referred to as a **Geiger counter**) tells how much radiation is given off by a source in 'real time'.

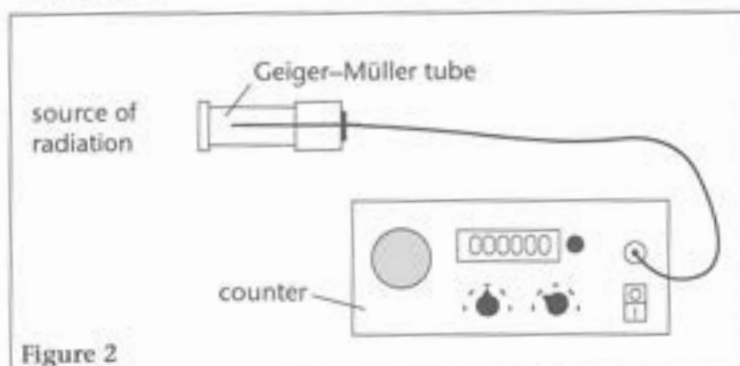


Figure 2

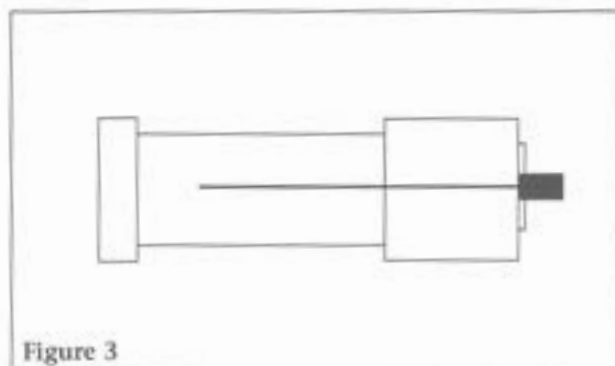


Figure 3

Figure 2 shows how a Geiger counter is set up to detect and measure radiation. Figure 3 is a simplified view of a Geiger-Müller tube. The tube consists of an outer casing with a 'window' at the front of the tube that allows radiation to pass through. The inside has a central wire at a potential of around + 400 volts surrounded by argon gas. When ionising radiation enters the tube, it strips an electron from an argon atom. The electron (having a negative charge) is attracted to the central wire. As it accelerates towards the positive charge, the electron collides with other argon atoms which causes more electrons to be released. The group of electrons arrives at the wire which causes a pulse of current that is detected, amplified and counted.

- 3 Copy figure 3. Use the information from the text to label the parts of the tube.
- 4 Outline the advantages and disadvantages of detecting radiation with a Geiger counter compared to a photographic film badge.

Sources of radioactivity

- 5 Some sources of radioactivity are natural and others are made by humans. Mark each source of radiation as natural or caused by humans.
 - a Radiation from outer space, e.g. cosmic rays
 - b Waste from nuclear power stations
 - c Radioisotopes deep in the Earth's core
 - d 'Fallout' from nuclear explosions

Half-life A source of radioactivity cannot give out energy and particles forever. As time goes by the amount emitted must decline. The concept of a 'half-life' is used to describe how long a substance will remain radioactive.

- 6 Use a textbook to give another definition of *half-life*.

The **half-life** of a radioisotope is the average time it takes for half of the radioactive atoms to decay to a different atom.

The half-life of a radioisotope varies from a fraction of a second to over a million years, depending on the particular radioisotope. Knowing the half-life of a radioisotope can help forecast how long it will remain a cause of concern. It can also help determine the age of artefacts. Iodine-131 has a half-life of 8 days. This means that after 8 days the radiation will be half of the original radioisotope; after 16 days, it will be a quarter; after 24 days, an eighth, and so on.

7 What percentage of the original iodine-131 isotope would be present after 40 days?

Carbon-14 is a naturally-occurring radioisotope present at constant levels in all living creatures. This radioisotope has a half-life of around 5 800 years. When the organism dies, the level of carbon-14 declines. By comparing the amount of carbon-14 present in an archaeological specimen with that in a living example, the age of the specimen can be calculated. This is called radiocarbon dating.

8 How much of the original carbon-14 will be in the remains of a tree after 11 600 years?

9 A skeleton has one-sixteenth of the carbon-14 of a living equivalent. How old is it likely to be?

10 A piece of pottery is thought to be around 10 000 years old. Would radiocarbon dating be useful in establishing how old this object is? Explain your answer.

Uses of radioisotopes

The use of a radioactive isotope depends on the penetrating power of its radiation and the length of its half-life. Some applications need a steady source of radiation (long half-life) and others need radiation that diminishes in strength quickly.

Alpha particle sources Alpha particles only penetrate weakly. Smoke detectors tend to use a source of alpha radiation that sends particles to a detector across a gap of air. When smoke particles are present, this reduces the number of alpha particles reaching the detector and electronic circuitry can generate an alarm.

Beta radiation sources Beta particles are usually stopped by a solid material a few centimetres thick. The amount of radiation absorbed by the material is proportional to the thickness (and density) of the material. The thicker the material, the more beta radiation it absorbs. Materials that are produced in sheet form to absorb certain amounts of radiation can have their thicknesses monitored. The sheet passes between a beta source and a detector. If the amount of radiation detected is too high, the sheet is too thin; if the radiation is too low, the sheet is too thick. This information is fed back to the production machinery to alter thickness.

Gamma radiation sources Gamma radiation is the most highly penetrating and is used when radiation must be able to pass through a substantial thickness of a material. It can be used as a tracer; e.g. to detect leaks in pipes carrying fluids and, after injection, to follow biological processes in the human body. For the latter use, the half-life should be as short as possible to minimise any health hazards. The destructive nature of gamma radiation can be used to kill cancerous cells (radiotherapy) and to sterilise food and surgical instruments. The gamma emitters in the instruments used to destroy cells and bacteria need to have relatively long half-lives to allow the instruments to work for a reasonable length of time.

11 Draw up a table to summarise the uses of radiation. Use the headings below.

Type of radiation	Function	Half-life	Penetrating power
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12 Give another way radiotherapy is used and briefly describe how it works.

13 Describe how technicians who work in radiotherapy clinics might be put at risk and how these risks can be minimised.

14 Match each term to the correct definition: *radioisotope*, *gamma radiation*, *beta radiation*, *isotope*, *alpha radiation*.

- a High-speed electrons given off during radioactive decay
- b High-energy (short-wavelength) radiation emitted during radioactive decay
- c A radioactive isotope
- d The nucleus of a helium atom, given off during radioactive decay
- e An element with the same number of protons but a different number of neutrons

The petrochemical industry is of crucial importance in modern life. It supplies a large portion of our energy needs, such as petrol, diesel fuel and methane. From crude oil (petroleum) it supplies the raw materials used in the manufacture of many consumer products. Crude oil is a complex mixture of hydrocarbons.

1 What is a hydrocarbon?

Crude oil is of little use in its raw form. It has to be treated before it is of commercial use. The first step is fractional distillation in a refinery. This separates it into various fractions.

2 What difference in physical properties is utilised in fractional distillation?

Each fraction contains a mixture of hydrocarbons with similar chain length.

3 What is meant by *similar chain length*?

The proportion of hydrocarbons present in crude oil does not correspond with society's demands. The chemicals present are modified to match supply with demand. The long chains of alkanes present can be broken to form shorter (and more useful) compounds.

4 What is an *alkane*?

Cracking When alkanes are heated, their molecules vibrate. At high temperatures they vibrate so strongly that carbon-carbon bonds are broken and smaller molecules are formed. This process is called **thermal cracking**. One of these smaller molecules is an alkene.

5 What is an *alkene*?

By using a catalyst, thermal cracking can be made to occur at lower temperatures in a process called **catalytic cracking**.

6 Define *catalyst*.

This is the cracking of a typical alkane, undecane to nonane and ethene (ethylene, an alkene):



7 Draw the structural formula for ethene.

8 What is the major structural difference between ethene and the two alkanes, undecane and nonane?

Because ethene has a double bond (**unsaturated**), it is a reactive compound. Ethene is the raw material in the formation of ethanol ($\text{C}_2\text{H}_5\text{OH}$), ethan-1, 2-diol (ethylene glycol, $\text{HOCH}_2\text{CH}_2\text{OH}$) and polythene.

9 Give one use for each compound.

- a ethanol
- b ethan-1, 2-diol
- c polythene

10 One way of forming ethanol is by reacting ethene with water using a catalyst at high temperature and pressure.

- a Write a balanced equation for this reaction.
- b Why do you think a catalyst was used?
- c Give the name of another process that produces ethanol.

11 Why is petroleum cracked?

12 Why is an alkene usually more reactive than the equivalent alkane?

13 Name three petrochemical fuels.

CORROSION OF IRON

When a metal is attacked by air, water or other substances, the metal is said to **corrode**. Usually the more reactive a metal is, the more readily it corrodes. The most common example of corrosion is the rusting of iron. Corrosion is an **oxidation** process. Remember, oxidation involves any (or all) of the following:

- addition of oxygen
- loss of electrons
- loss of hydrogen.

1 *Reduction* is the opposite of oxidation. Outline what reduction involves.

Corrosion is a process that costs Australia billions of dollars each year. Steel, an alloy of iron, is a commonly used structural material; it is strong, easily shaped and reasonably cheap. However, it rusts in damp air.

2 Define *alloy*.

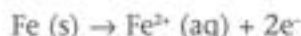
3 Give two structural uses of steel.

When iron corrodes, rust forms in flakes. Damp air can get past the flakes and attack the metal below. Given enough time, a piece of iron can rust all the way through.

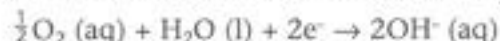
There are many ways to protect iron from rusting. They often involve coating the iron with paint or grease to keep out air and water. **Galvanising** is a process that covers iron (or steel) with a coating of zinc. Zinc is more reactive than iron, so will corrode in preference to the iron even if the zinc coating is scratched. Rust is hydrated iron oxide and has the formula $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$, where x can be a number between one and three.

4 What is the oxidation state of iron in hydrated iron oxide?

Rust forms in a series of reactions. The first step is the oxidation of iron:



The electrons lost in this reaction are used by oxygen (which is reduced):



5 Add the two previous half-reactions to show the formation of iron (II) hydroxide, Fe(OH)_2 .

The iron (II) hydroxide formed is insoluble. This in turn reacts with further oxygen and moisture to form iron (III) hydroxide, a reddish-brown solid.

6 Copy and balance the equation below which shows the process described above.



In air, the iron (III) hydroxide dehydrates to form rust: $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$.

7 What is meant by *dehydrates*?

The wet corrosion of iron involves the setting up of electrochemical cells on the surface of the iron, where **anodes** and **cathodes** are formed.

An anode:

- is where iron loses electrons and is oxidised
- has lower oxygen concentration than the cathode.

8 A *cathode* can be regarded as the opposite of an anode. Give the characteristics of an anode.

9 Give three desirable properties of steel.

10 Give an undesirable property of steel.

11 Define *corrode*.

12 Why can coating iron with paint stop (or slow) corrosion?

13 Define *galvanising*.

SELECTED ANSWERS 1–7

1 Polymers 1

1 Teacher to check; should mention the size and repetition of monomers 2 $C_6H_{12}O_6$ 3 Literally, many sugars 4 A large number 5 e.g. rice, potatoes, wheat 6 Polymers made up of glucose, starch edible by humans, cellulose inedible/stiffness 7 e.g. Clothing, parachutes 8 e.g. Wool 9 e.g. Tyres, waterproof clothing 10 C_4H_7 11 Softens when hot, brittle when cold and perishes over time 12 The addition of sulfur to rubber to improve its properties 13 Glucose 14 Can't see through it 15 Any plant 16 e.g. Billiard balls, ornaments and (white) piano keys 17 e.g. It's not ethical to use ivory, a substitute could be cheaper and more available 18 e.g. Flexible, transparent, able to be have an image imposed on it 19 Cocoon of silkworm larvae. Silk worms must be destroyed before they are ready to break the cocoon. This way the single thread used to make it is in one piece, then it can be unwound and woven. 20 To obtain characteristics of different fibres, reduce cost 21 e.g. Clay, copper 22 Teacher to check; should include the change of shape 23 phenol: C_6H_5OH ; methanal: $HCOH$ 24 e.g. Tough, heat resistant, cheap

2 Polymers 2

1 When you want a plastic item to keep its shape when exposed to hot foods or liquids 2 When you want to be able to recycle a plastic item by melting it 3 C_2H_4 4 Repeated C_2H_4 units, n times, n being a large number 5 Large molecules made up of many repeating subunits called monomers 6 Softens when heated and hardens on cooling 7 Carbon-carbon double bond 8 Very cheap, strong, flexible 9 They don't biodegrade, manufacture uses non-renewable resources i.e. petrochemicals 10 C_2H_3Cl 11 $(C_2H_3Cl)_n$ 12 Too large to show it all 13 Propylene/propene 14 Teacher to check; should be the same as ethylene, but with an H replaced by CH_3 15 C_3H_6 16 Thermoplastic – very similar to polyethylene 17 Chemicals obtained from crude oil, natural gas and coal 18 C_8H_8 19 Low density, can float 20 Teacher to check; should be similar to ethylene with the H replaced by F 21 Teacher to check; should be similar to polyethylene with the H replaced by F 22 Homopolymer only has one type of monomer, a copolymer has two or more 23 Hydrocarbon with two carbon-carbon double bonds 24 General, non-specific 25 Carbon, hydrogen, nitrogen and oxygen 26 Addition is carbon-carbon double bonds breaking to form single bonds with other molecules; condensation has a molecule of water eliminated 27 Natural product became scarce 28 Teacher to check

3 Problems with plastics?

1 Low-density polyethylene 2 e.g. Cost, to reduce waste/litter 3 Paper biodegrades, is made from renewable materials; not as strong and tears when wet 4 People need to return to the store to recycle, bags may be of different materials and need to be sorted, bags may be dirty 5 Danger of suffocation of young children 6 Teacher to check; should include less use of raw materials and energy, less waste, hard to recycle, little profit 7 No, not exposed to light 8 Biodegradable polymer can be broken down by the action of organisms,

biopolymers are made by organisms 9 Teacher to check; should include the dot points at the beginning of the sheet 10 Teacher to check that the substance is a starch

4 The carbon cycle

1 Respiration, decay, burning 2 Photosynthesis 3 e.g. Production of a carbohydrate (and oxygen) by a plant using water and carbon dioxide and sunlight as energy 4 Catalyst 5 Carbon dioxide and water 6 One-tenth of a percent, 0.1% 7 Millions of years 8 $C_6H_{12}O_6 + 6O_2 \rightarrow 6CO_2 + 6H_2O$ 9 Energy (heat) given out 10 e.g. Coal, oil, natural gas 11 Increase greenhouse effect, contribute to global warming 12 Cellulose, starch 13 Decay, release materials back into the cycle 14 a True b True c True d True 15 Teacher to check. Include: oceans (and seas) have huge amounts of carbon dioxide dissolved. Some organisms use carbon dioxide in photosynthesis (Phytoplankton), some carbon dioxide is taken up by coral, and some carbon dioxide is generated by normal respiration.

5 Carbohydrates

1 $6CO_2(g) + 6H_2O(l) \rightarrow C_6H_{12}O_6(s)$ 2 Endothermic, needs energy from the sun 3 Teacher to check 4 $C_6H_{12}O_6 + C_6H_{12}O_6 \rightarrow C_{12}H_{22}O_{11} + H_2O$ 5 Glucose is a six-sided ring, fructose is five-sided, arrangement of Hs and -OH groups 6 Made up of two 'single' sugars, i.e. two monosaccharides 7 A molecule of water is formed. 8 Sugar with six carbon atoms per molecule 9 Could include: all contain glucose as a monomer, cellulose and starch are from plants, glycogen in animals, only glycogen is soluble in water. 10 It gives a blue/black colour when added to iodine solution. 11 Cellulose is renewable. 12 Copper (II), Cu^{2+}

6 Amino acids and proteins

1 Carbohydrate, fat 2 Make them soluble, water is a polar solvent 3 Amino acid from $-NH_2$ (amine) and acid from $-COOH$ (carboxylic acid) 4 Teacher to check; refer to figure 2 5 Teacher to check; refer to figure 2 6 A larger molecule made up of peptides 7 Children are growing more than adults are. 8 Covalent, non-metal to non-metal 9 Teacher to check 10 Teacher to check 11 Carbonyl 12 Amine 13 Teacher to check 14 Primary: amino acid sequence; secondary: 3-D shape; tertiary: distortions of 3-D shape. 15 Glycine and alanine, see figure 2 16 Teacher to check 17 Both speed up reactions. Enzymes are more specific than inorganic catalysts, work under milder conditions and tend to operate at lower temperatures.

7 Fermentation

1 $C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$ 2 Carbon dioxide being lost from the system 3 Does it automatically, can take readings outside of school time 4 There were 5×24 readings, i.e. too many to write down, only a representative sample shown 5 Mass lost, in table order: 0, 1.2, 2.4, 3.2, 3.8, 4.1, 4.3, 4.4 ... 6 Teacher to check 7 Mass 8 Time 9 Approximately 30 hours after start. Mass remained constant, CO_2 no longer being given off 10 10.0 g 11 180.2 g 12 0.0555 13 4.4 g 14 44.01 g 15 0.01 16 0.24 L

SELECTED ANSWERS 7–12

17 Teacher to check; e.g. ensure only temperature was changed, new temperature kept steady 18 Anaerobic enzymatic conversion of organic compounds to alcohols 19 e.g. Beer, spirits, bread

8 Ethanol and ethene

1 $\text{CH}_3\text{CH}_2\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 3\text{H}_2\text{O}(\text{l}) + 2\text{CO}_2(\text{g})$
 2 Left, ethanol, $\text{C}_2\text{H}_6\text{O}$; right ethene, C_2H_4 3 1, 2-ethandiol, $\text{C}_2\text{H}_4\text{O}_2$ 4 $\text{C}_6\text{H}_{12}\text{O}_6(\text{aq}) \rightarrow 2\text{CH}_3\text{CH}_2\text{OH}(\text{aq}) + 2\text{CO}_2(\text{g})$ 5 $\text{CH}_2\text{CH}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{CH}_3\text{CH}_2\text{OH}(\text{l})$
 6 $\text{CH}_3\text{CH}_3(\text{g}) \rightarrow \text{CH}_2\text{CH}_2(\text{g}) + \text{H}_2(\text{g})$ 7 Solvent, fuel, raw material for other chemicals 8 Making polythene, 1, 2-ethandiol and ethanol 9 Carbohydrate, water and yeast 10 a Can mix with b Mixture of petrol (gasoline) and ethanol c No oxygen present d Addition of water e Organic catalyst f Breaking hydrocarbon chain g Anaerobic enzymatic conversion of organic compounds to alcohols 11 H_3PO_4
 12 When ethanol percentage gets too high, it kills the yeast. 13 Fractional distillation

9 Esters

1 $\text{HCOOH} + \text{CH}_3\text{OH} \rightleftharpoons \text{HCOOCH}_3 + \text{H}_2\text{O}$
 2 $\text{CH}_3\text{COOH} + \text{CH}_3\text{CH}_2\text{OH} \rightleftharpoons \text{CH}_3\text{COOCH}_2\text{CH}_3 + \text{H}_2\text{O}$
 3 Pentyl ethanoate, $\text{C}_2\text{H}_3\text{OOC}_5\text{H}_{11}$, HCOOC_2H_5 , Methanoic, Ethanol, $\text{C}_2\text{H}_3\text{OOC}_8\text{H}_{17}$, Ethanoic, Octanol
 4 Parts per billion 5 One where water is formed
 6 Smell, perfume 7 Yes, because flowers and fruits often form the basis of perfumes

10 The atmosphere 1

1 Troposphere, 0 to 12, Almost all weather occurs here
 Tropopause, 12 to 16, Transition zone
 Stratosphere, 16 to 50, Only minor weather changes
 Ozonosphere, 16 to 50 (approx.), Ozone produced here
 Mesosphere, 50 to 80, Temperature drops
 Ionosphere, 50 upwards, Contains charged particles
 Thermosphere, 80 to 450, Increase in temperature
 Exosphere, 450 upwards, Outer edge of atmosphere.
 2 Teacher to check; students should observe things like pressure changes noticeably when a change of weather comes
 3 Nitrogen, N_2 , Element, 70.08
 Oxygen, O_2 , Element, 20.95
 Argon, Ar, Element, 0.94
 Water (vapour), H_2O , Compound, Variable
 Carbon dioxide, CO_2 , Compound, Minute
 Neon, Ne, Element, Minute
 Helium, He, Element, Minute
 Methane, CH_4 , Compound, Minute
 Krypton, Kr, Element, Minute
 Ozone, O_3 , Element, Minute
 Sulfur dioxide, SO_2 , Compound, Minute
 4 e.g. Gradual change, arbitrary areas, defined by temperature changes 5 Teacher to check; should include description of Greenhouse effect and the role of gases 6 Add 273. 7 Nitrogen first, then argon and then oxygen 8 Molecule made of two atoms; e.g. N_2
 9 Teacher to check; suggestions include – Daniel Rutherford: Medical student from Edinburgh. Joseph Priestley: 1733–1804, invented soda water, discovered many other gases, (wrongly) supported phlogiston

theory. Henry Cavendish: 1731–1810, investigated hydrogen, specific heat. Antoine Lavoisier: 1743–1794, explained combustion, named oxygen 10 Maintain the properties of chips for longer in storage – chips won't oxidise 11 Flash-freezing food, preserving biological specimens 12 $\text{N}_2(\text{g}) + 3\text{H}_2(\text{g}) \rightarrow 2\text{NH}_3(\text{g})$
 13 Teacher to check; process of incorporating atmospheric nitrogen into plant material 14 Legumes are any members of the pea family 15 $2\text{HgO}(\text{s}) \rightarrow 2\text{Hg}(\text{l}) + \text{O}_2(\text{g})$ 16 $\text{C}_2\text{H}_2(\text{g}) + 2\frac{1}{2}\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + \text{H}_2\text{O}(\text{l})$
 17 Carbon 12

11 The atmosphere 2

1 If in air (containing oxygen), the tungsten metal wire would eventually oxidise. The argon is unreactive and the tungsten lasts much longer. It also stops the tungsten evaporating, which would happen if the inside of the globe were a vacuum. 2 Elements exist as single atoms. 3 Differences in boiling points 4 Teacher to check; should agree with text 5 The science that deals with reaching low temperatures and the behaviour of substances at these temperatures 6 W 7 Ra
 8 Have full outer (valence) electron shells 9 $\text{CO}_2(\text{g}) + \text{Ca}(\text{OH})_2(\text{aq}) \rightarrow \text{CaCO}_3(\text{s}) + \text{H}_2\text{O}(\text{l})$ 10 Teacher to check; could include that it is natural and is not a pollutant, could also state that increased human activities have caused it to become a pollutant 11 a False b False c False 12 Substance (not usually in place) that causes harm 13 $\text{C}_8\text{H}_{18}(\text{l}) + 12\frac{1}{2}\text{O}_2(\text{g}) \rightarrow 8\text{CO}_2(\text{g}) + 9\text{H}_2\text{O}(\text{l})$ 14 C_6H_6 15 Evaporation 16 Toxic and carcinogenic 17 Vehicles produce CO 18 $\text{NO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{HNO}_3(\text{aq})$ 19 Harmful to breathing systems, forms ground-level ozone and smog 20 $\text{SO}_2(\text{g}) + \text{H}_2\text{O}(\text{l}) \rightarrow \text{H}_2\text{SO}_3(\text{aq})$ 21 Carbon dioxide or an oxide of nitrogen 22 Sulfur dioxide (or oxides of nitrogen) in the atmosphere, usually from mining and burning, dissolve in water droplets and fall as highly acidic rain. Carbon dioxide also plays a part. 23 Oxygen 24 In the ozone layer, screens out harmful UV rays 25 Irritates eyes and damages the respiratory system when at ground level 26 Teacher to check; include propellant for aerosols, damaging to the ozone layer 27 In the form of lead tetraethyl, it helped improve the 'octane rating', i.e. engines ran better, but it is poisonous. 28 Teacher to check

12 Roles of chemists

1 matter Anything that has mass and takes up space; petrochemical to do with chemical obtained from petroleum; pharmaceutical drugs of a medicinal nature; quality control ensuring items are up to standard; research and development developing new materials and processes; raw materials materials used to make products; analysis study, margins limits; emissions substances given out; products things made; automated operated by machines; commercial financially profitable; academic institutions places where teaching and learning take place on an organised basis; chemistry science that deals with structure and reactions of substances; biochemistry the chemistry of living things; forensic science applied to legal matters 2 If something goes wrong at one step the final product is faulty. 3 Very expensive 4 Later, developments can be matched to needs. 5 Teacher to check

SELECTED ANSWERS 13–20

13 Sulfur

1 Brittle, doesn't conduct electricity, low melting point
 2 Oxygen, selenium, tellurium 3 16 4 Compounds based on the element carbon 5 Teacher to check
 6 SO_3 7 Sulfa drugs, matches, rubber tyres, gunpowder, sulfuric acid, paints, pigments 8 Acts as a preservative by killing germs (bacteria) 9 Power generation, metal production 10 Lead, sulfuric acid, water 11 Making organic products, paints, artificial fabrics, treating metals, car batteries 12 Element that exists in more than one physical form 13 Teacher to check; should label flow of steam, air and sulfur/steam/air mixture 14 Low melting point and its insolubility in water 15 $\text{S} (\text{s}) + \text{O}_2 (\text{g}) \rightarrow \text{SO}_2 (\text{g})$ 16 $4\text{FeS} (\text{s}) + 7\text{O}_2 (\text{g}) \rightarrow 2\text{Fe}_2\text{O}_3 (\text{s}) + 4\text{SO}_2 (\text{g})$ 17 $\text{SO}_2 (\text{g}) + \frac{1}{2}\text{O}_2 (\text{g}) \rightarrow \text{SO}_3 (\text{g})$ 18 $\text{H}_2\text{SO}_4 (\text{l}) + \text{SO}_3 (\text{g}) \rightarrow \text{H}_2\text{S}_2\text{O}_7 (\text{l})$ 19 $\text{H}_2\text{S}_2\text{O}_7 (\text{l}) + \text{H}_2\text{O} (\text{l}) \rightarrow 2\text{H}_2\text{SO}_4 (\text{l})$ 20 Pollution, cost, loss of raw materials 21 Making fertilisers 22 Teacher to check; should follow stages in text 23 a False b False c True d True

14 Iron occurrence and manufacture

1 56 amu, 26 2 'In between' elements; transition elements/metals are found between periods II and III
 3 $\text{C} (\text{s}) + \text{O}_2 (\text{g}) \rightarrow \text{CO}_2 (\text{g})$ 4 $\text{C} (\text{s}) + \text{CO}_2 (\text{g}) \rightarrow 2\text{CO} (\text{g})$ 5 $3\text{CO} (\text{g}) + \text{Fe}_2\text{O}_3 \rightarrow 2\text{Fe} (\text{s}) + 3\text{CO}_2 (\text{g})$
 6 $\text{CaCO}_3 (\text{s}) \rightarrow \text{CaO} (\text{s}) + \text{CO}_2 (\text{g})$ 7 $\text{CaO} (\text{s}) + \text{SiO}_2 (\text{s}) \rightarrow \text{CaSiO}_3 (\text{s})$ 8 $\text{CaO} (\text{s}) + \text{Al}_2\text{O}_3 (\text{s}) \rightarrow \text{CaAl}_2\text{O}_4 (\text{s})$
 9 a Uncombined with other elements b Plentiful, common c High melting point d Heat taken in
 10 Has more than one stable oxidation state in its compounds, can act as a catalyst, forms a variety of complex ions 11 Making cement, roads and building materials 12 Impure iron from a blast furnace

15 The Solvay process

1 $\text{CaCO}_3 (\text{s}) \rightarrow \text{CaO} (\text{s}) + \text{CO}_2 (\text{g})$ 2 $\text{NH}_3 (\text{g}) + \text{CO}_2 (\text{g}) + \text{NaCl} (\text{aq}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{NH}_4\text{Cl} (\text{aq}) + \text{NaHCO}_3 (\text{s})$
 3 $2\text{NaHCO}_3 (\text{s}) \rightarrow \text{Na}_2\text{CO}_3 (\text{s}) + \text{CO}_2 (\text{g}) + \text{H}_2\text{O} (\text{l})$
 4 $\text{CaO} (\text{s}) + \text{H}_2\text{O} (\text{l}) \rightarrow \text{Ca}(\text{OH})_2 (\text{aq})$ 5 $2\text{NH}_4\text{Cl} (\text{aq}) + \text{Ca}(\text{OH})_2 (\text{aq}) \rightarrow 2\text{NH}_3 (\text{g}) + 2\text{H}_2\text{O} (\text{l}) + \text{CaCl}_2 (\text{aq})$
 6 $2\text{NaCl} (\text{aq}) + \text{CaCO}_3 (\text{s}) \rightarrow \text{Na}_2\text{CO}_3 (\text{s}) + \text{CaCl}_2 (\text{aq})$
 7 Teacher to check 8 Sodium chloride and calcium chloride 9 To soften 'hard' water, and to make soap and glass

16 Redox 1

1 Valency 2 a Transfer of electrons b Sharing electrons c Unequal sharing of electrons 3 If an electron is 'given', then there must be an atom that can 'accept' it at the same time. 4 Oxidation
 5 a Oxidising agent or oxidant b Reducing agent or reductant 6 a Magnesium oxidised, oxygen reduced b Zinc oxidised, sulfur reduced c Iron oxidised, copper ion reduced 7 The halogens, Group VII 8 The alkali metals, Group I 9 e.g. Burning, corrosion, respiration, photosynthesis 10 a -3 b +5 c -2 d +6

17 Redox 2

1 +6 2 a Chromium (II) oxide (chromous oxide) b Chromium (III) oxide (chromic oxide) c Chromium (VI) oxide (chromium trioxide) 3 a Fe_2O_3 b $\text{Co}(\text{OH})_2$ 4 a Na from 0 to +1, oxidised; Cu from +2 to 0, reduced;

Cl unchanged on -1 b Zn from 0 to +2, oxidised; H from +1 to 0, reduced; Cl unchanged on -1 c Si from +2 to 0, reduced; Mg to +2, oxidised; O unchanged on -2 d No changes e Cl from +5 to -1, reduced, O from -2 to 0, oxidised 5 a $\text{Cu} (\text{s}) + 2\text{AgNO}_3 (\text{aq}) \rightarrow \text{Cu}(\text{NO}_3)_2 (\text{aq}) + 2\text{Ag} (\text{s})$ or $\text{Cu} (\text{s}) + 2\text{Ag}^+ (\text{aq}) \rightarrow \text{Cu}^{2+} (\text{aq}) + 2\text{Ag} (\text{s})$ b $\text{TiCl}_4 (\text{s}) + 4\text{Na} (\text{s}) \rightarrow \text{Ti} (\text{s}) + 4\text{NaCl} (\text{s})$ c $\text{Pb} (\text{s}) + 2\text{HCl} (\text{aq}) \rightarrow \text{PbCl}_2 (\text{s}) + \text{H}_2 (\text{g})$ 6 Cu from +2 to 0, reduced; C from 0 to +4, oxidised 7 a $\text{SnO}_2 (\text{s}) + \text{C} (\text{s}) \rightarrow \text{Sn} (\text{s}) + \text{CO}_2 (\text{g})$ b +4, -2, 0 $\rightarrow$ 0, +4, -2 8 $\text{C} (\text{s}) + \frac{1}{2}\text{O}_2 (\text{g}) \rightarrow \text{CO} (\text{g})$ 9 a $3\text{CO} (\text{g}) + \text{Fe}_2\text{O}_3 (\text{s}) \rightarrow 3\text{CO}_2 (\text{g}) + 2\text{Fe} (\text{s})$ b +4, -4, +3, -2 $\rightarrow$ +4, -2, 0 10 $2\text{Al} (\text{s}) + \text{Fe}_2\text{O}_3 (\text{s}) \rightarrow \text{Al}_2\text{O}_3 (\text{s}) + 2\text{Fe} (\text{s})$; 0, +3, -2 $\rightarrow$ +3, -2, 0 11 $\text{CuO} (\text{s}) + \text{H}_2 (\text{g}) \rightarrow \text{Cu} (\text{s}) + \text{H}_2\text{O} (\text{l})$ 12 $\text{CH}_4 (\text{g}) + 2\text{PbO}_2 (\text{s}) \rightarrow 2\text{Pb} (\text{s}) + \text{CO}_2 (\text{g}) + 2\text{H}_2\text{O} (\text{l})$ 13 $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ 14 $\text{O}^{2-} + 2\text{e}^- \rightarrow \frac{1}{2}\text{O}_2$ 15 $2\text{Al}^{3+} + 3\text{O}^{2-} \rightarrow 2\text{Al} + 1\frac{1}{2}\text{O}_2$

18 Redox 3

1 +6 2 +6 3 +7 4 $\text{CH}_3\text{CH}_2\text{OH}$ or $\text{C}_2\text{H}_5\text{OH}$ 5 CH_3COOH , or $\text{C}_2\text{H}_4\text{O}_2$ 6 Reduced, oxidation number goes from VI to III 7 +1, +6, -2, +1, -1, 0, +4, -2, +1, -2 8 0, +1, +5, -2, +2, +5, -2, +4, -2, +1, -2 9 8 10 5 11 14 12 6 13 a $\text{Pb} (\text{s}) + \text{PbO}_2 (\text{s}) + 2\text{H}_2\text{SO}_4 (\text{aq}) \rightarrow 2\text{PbSO}_4 (\text{s}) + 2\text{H}_2\text{O} (\text{l})$; 0, +4, -2, +1, +6, -2, +2, +6, -2, +1, -2 b $\text{K}_2\text{Cr}_2\text{O}_7 (\text{aq}) + 14\text{HCl} (\text{aq}) \rightarrow 3\text{Cl}_2 (\text{g}) + 7\text{H}_2\text{O} (\text{l}) + 2\text{CrCl}_3 (\text{aq}) + 2\text{KCl} (\text{aq})$; +1, +6, -2, +1, -1 0, +1, -2, +3, -1, +1, -1 14 Zinc is negative, giving off electrons 15 Teacher to check; draw in pencil, using ruler, label in blue/black ink 16 Teacher to check 17 Allows ions to move (and hence completes the circuit) but minimises contamination of the two solutions 18 Potassium nitrate or ammonium chloride 19 $\text{Zn} (\text{s}) \rightarrow \text{Zn}^{2+} (\text{aq}) + 2\text{e}^-$, oxidation 20 $\text{Ag}^+ (\text{aq}) + \text{e}^- \rightarrow \text{Ag} (\text{s})$, reduction 21 $\text{Zn} (\text{s}) + 2\text{Ag}^+ (\text{aq}) \rightarrow \text{Zn}^{2+} (\text{aq}) + 2\text{Ag} (\text{s})$ 22 a Silver electrode b Zinc electrode

19 Redox 4

1 $\text{Zn} (\text{s}) \rightarrow \text{Zn}^{2+} (\text{aq}) + 2\text{e}^-$ 2 $\text{Ag} (\text{s}) \rightarrow \text{Ag}^+ (\text{aq}) + \text{e}^-$ 3 Teacher to check; draw in pencil, using ruler, label in blue/black ink 4 a Would decrease, zinc metal going into solutions as ions b Would increase, silver metal being deposited 5 a Magnesium metal forms (dipositive) magnesium ions and (dipositive) iron ions form iron metal; magnesium electrode is negative and iron electrode is positive b Nickel metal forms (dipositive) nickel ions and (dipositive) copper ions form copper metal; nickel electrode is negative and copper electrode is positive 6 a Galvanic b Oxidation c Electrode d Connected e Electrons f Salt g Charge h Positive i Electropositive 7 a $\text{Br}^+ + 2\text{e}^- \rightarrow 2\text{Br}^-$ b $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}$ c $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$ d $\text{I}_2 + 2\text{e}^- \rightarrow 2\text{I}^-$ 8 Teacher to check 9 No, E° value is negative, more likely to go right to left 10 a Voltmeter b Anode c Cathode d Salt bridge e Electrolyte

20 Acids and bases 1

1 Indicators; because they indicate whether something is an acid or a base 2 $\text{C}_6\text{H}_8\text{O}_7$ 3 Ethanoic acid, acetic acid or vinegar; citric acid from citrus fruits; lactic acid from sour milk 4 $2\text{HCl} + 2\text{Hg} \rightarrow \text{Hg}_2\text{Cl}_2 + \text{H}_2$ 5 No; teacher to check reasoning; e.g. mention substances that contain hydrogen but aren't acids 6 Acids, bases (alkalis), neutral 7 Lavoisier 8 Davy

SELECTED ANSWERS 20–26

(but there were others) **9** Base **10** H^+ **11** OH^-
12 a Hydrogen nitrate (nitric acid) + water $\rightarrow$ hydrogen ions + nitrate ions **b** Sodium hydroxide (base) + water $\rightarrow$ sodium ions + hydroxide ions **c** Ethanoic/ acetic acid + water $\rightarrow$ ethanoate/acetate ions + hydrogen ions
d Ammonium hydroxide (base) + water $\rightarrow$ ammonium ions + hydroxide ions **13 a** Water has transferred a proton to ammonia, forming ammonium and hydroxide ions. Water is the acid and ammonia is the base.
b Nitric acid has transferred a proton to water, forming nitrate and hydronium ions. Nitric acid is the acid and water is the base. **c** Water has transferred a proton to sodium hydroxide, forming sodium and hydroxide ions. Water is the acid and sodium hydroxide is the base.
14 a Bases **b** Acids **c** Acids and bases **d** Bases
e Acids **f** Bases **g** Acids **15** Acid is an electron acceptor, base is an electron donor **16** Hydrobromic, Bromides; Nitric, Nitrates; H_2SO_4 , Sulfates; Ethanoic/ acetic, CH_3COOH **17** An ionic substance that is formed when a base is neutralised by an acid

21 Acids and bases 2

1 a $\text{HCl (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{H}_3\text{O}^+ \text{ (aq)} + \text{Cl}^- \text{ (aq)}$ **b** $\text{HBr (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{H}_3\text{O}^+ \text{ (aq)} + \text{Br}^- \text{ (aq)}$ **c** $\text{H}_2\text{SO}_4 \text{ (aq)} + 2\text{H}_2\text{O (l)} \rightleftharpoons 2\text{H}_3\text{O}^+ \text{ (aq)} + \text{SO}_4^{2-} \text{ (aq)}$ **2** CH_2O_2
3 a $\text{CH}_3\text{COOH (l)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{H}_3\text{O}^+ \text{ (aq)} + \text{CH}_3\text{COO}^- \text{ (aq)}$
b $\text{HF (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{H}_3\text{O}^+ \text{ (aq)} + \text{F}^- \text{ (aq)}$ **c** $\text{H}_2\text{CO}_3 \text{ (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{H}_3\text{O}^+ \text{ (aq)} + \text{HCO}_3^- \text{ (aq)}$ or $\text{H}_2\text{CO}_3 \text{ (aq)} + 2\text{H}_2\text{O (l)} \rightleftharpoons 2\text{H}_3\text{O}^+ \text{ (aq)} + \text{CO}_3^{2-} \text{ (aq)}$ **4 a** $\text{NaOH (s)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{Na}^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$ **b** $\text{KOH (s)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{K}^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$ **c** $\text{NH}_3 \text{ (g)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{NH}_4^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$ or $\text{NH}_4\text{OH (aq)} + \text{H}_2\text{O (l)} \rightleftharpoons \text{NH}_4^+ \text{ (aq)} + \text{OH}^- \text{ (aq)}$
5 Teacher to check **6** Completely, partially, bases, strong, equilibrium **7 a** $\text{HNO}_3\text{--NO}_3^-$ and $\text{H}_3\text{O}^+\text{--H}_2\text{O}$
b HBr--Br^- and $\text{H}_3\text{O}^+\text{--H}_2\text{O}$ **c** $\text{H}_2\text{SO}_4\text{--SO}_4^{2-}$ and $\text{H}_3\text{O}^+\text{--H}_2\text{O}$
d $\text{C}_6\text{H}_5\text{NH}_2\text{--C}_6\text{H}_5\text{NH}_3^+$ and $\text{H}_2\text{O--OH}^-$ **e** $\text{N}_2\text{H}_4\text{--N}_2\text{H}_5^+$ and $\text{H}_2\text{O--OH}^-$ **f** $\text{NH}_3\text{--NH}_4^+$ and $\text{H}_2\text{O--OH}^-$
8 a Accepting/ receiving a proton **b** Donating a proton
9 Generates two protons per molecule **10** Generates three protons per molecule **11 a** False **b** False
c True **d** False **e** True **f** True **g** False **h** True
12 a Break up and form charged particles **b** Acids that generate more than one proton per molecule **c** Acids in which only small amounts ionise in water are weak. Strong acids ionise completely or almost completely.
d Balanced state, forward reaction equal to reverse reaction **e** Species that can donate and/or accept protons

22 Acids and bases 3

1 a 1.33 **b** 2.92 **c** 1.37 **d** 1.82 **e** 0.12 **2** Teacher to check; should agree with text **3** Acidic, acidic, neutral, basic (alkaline), basic (alkaline) **4** Change the sign of the power of ten to get pH **5 a** 0.40 **b** 5
c -0.50. Yes, you can get negative pH numbers, but these will usually be shown as zero on pH meters and indicator charts. **d** 2.5 **e** 6.0 **f** 11.5 **6** The best/most you can get **7** Organisms living in the sea
8 The negative log (to the base 10) of the hydrogen ion concentration expressed in moles per litre **9** Slightly acidic, hydrogen ion concentration is $1 \times 10^{-3.3}$
10 Teacher to check; could include methyl orange, red to yellow at pH 3.1 to 4.4; bromothymol blue, yellow to blue at pH 6.2 to 7.6; phenolphthalein, colourless to pink, at pH 8.3 to 10.0 **11** Descriptions will vary but typical answers include: pHs 0 to 1 red; 2, orange-red; 3,

orange; 4, pale orange; 5, orange-yellow; 6, yellow; 7, yellow-green, 8, green; 9, dark green; 10 to 14, blue
12 Advantages: more accurate, connect to data loggers; disadvantages: expensive, more complex to use
13 Solutions of either a weak acid and its salt, or a weak base and its salt; they resist changes in pH.

23 Nuclear chemistry 1

1 One ten-thousandth, 0.0001 **2** Electron, negative; 1.00 amu, nucleus; Proton, positive **3** Two **4** 35 amu
5 35.5 amu **6** 63.6 amu **7 a** Neutron **b** Mass spectrograph **c** Isotope **d** Amu **8** Teacher to check, but should agree with text. **9** Teacher to check, but should agree with text **10** When they hit particles, can change them to ions **11** Very high ionising power, large mass and when in the body, doesn't have to travel far to cause cell damage.

24 Nuclear chemistry 2

1 Teacher to check **2** The more the film has darkened, the greater the exposure to radiation **3** Teacher to check **4** Advantages: immediate check, counts actual numbers; disadvantage: more costly than film badge
5 a Natural **b** Human **c** Natural **d** Human
6 Teacher to check; e.g. time for radiation to drop to half of its initial value **7** 3.13% **8** 25% **9** 23 200 years
10 No, not organic **11** Teacher to check **12** Teacher to check **13** e.g. Always handling radioactive substances, use lead shields between patient and technician
14 a Beta radiation **b** Gamma radiation
c Radioisotope **d** Alpha radiation **e** Isotope

25 Petrochemicals: review

1 Compound that contains only hydrogen and carbon
2 Boiling point **3** Similar number of carbon atoms
4 Hydrocarbon with carbon-carbon single bonds only
5 Hydrocarbon with carbon-carbon double bonds
6 Substance that speeds up a reaction without being used up itself **7** Teacher to check **8** It is 'unsaturated', has a carbon-carbon double bond **9 a** Fuel, beverage, raw material **b** 'Antifreeze' **c** Making 'plastic' objects
10 a $\text{C}_2\text{H}_4 \text{ (g)} + \text{H}_2\text{O (l)} \rightarrow \text{C}_2\text{H}_5\text{OH (l)}$ **b** Less severe conditions, faster **c** Fermentation **11** To break it into shorter, more useful molecules **12** Alkenes have reactive carbon-carbon double bonds **13** Petrol, methane, diesel fuel

26 Corrosion of iron

1 e.g. Loss of oxygen, gain of electrons, gain of hydrogen
2 Mixture of two or more metals **3** Making frames for buildings, ships **4** +3 **5** $\text{Fe (s)} + \frac{1}{2}\text{O}_2 \text{ (aq)} + \text{H}_2\text{O (l)} \rightarrow \text{Fe}^{2+} \text{ (aq)} + 2\text{OH}^- \text{ (aq)}$ **6** $2\text{Fe(OH)}_2 \text{ (s)} + \frac{1}{2}\text{O}_2 \text{ (aq)} + \text{H}_2\text{O (l)} \rightarrow 2\text{Fe(OH)}_3 \text{ (s)}$ **7** Loses water **8** It is where oxygen gains electrons and is reduced, has higher oxygen concentration than the anode **9** Strong, cheap and is shaped easily **10** It rusts in damp air. **11** When a metal is attacked by air, water (or other substances) the metal is said to corrode. **12** Stops air and water from coming into contact with iron **13** Coating with zinc to prevent corrosion

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Extension work for advanced students	✓	A way of targeting student weaknesses	✓
Preparation for assessment	✓	Suitable for individual or group work	✓
Exam revision	✓	A variety of activities for home study	✓

Use them to supplement your existing resources, to extend student capabilities, for revision purposes, or for class work when the regular teacher is absent.

Assignments in Chemistry Book 3

This new three-book chemistry series by highly-experienced science authors and teachers Peter Gribben and Wendy St Germain provides photocopiable worksheets designed to challenge and stimulate students, as they explore a variety of chemistry concepts.

There are an array of opportunities for students to test their scientific skills through the analysis and interpretation of diagrams, tables and graphs. Students will also extend their reasoning and literacy skills as they are guided through the application of theoretical concepts to practical examples.

Research tasks are included to expand students' knowledge of the history of chemistry and the contributions of individual scientists.

Each assignment is self-contained and requires minimal teacher preparation. Answers have been provided for ease of marking.

Topics include

Polymers
The carbon cycle
Carbohydrates
Amino acids and proteins
Fermentation
Ethanol and ethene
The atmosphere
Roles of chemists

The Frasch process
The occurrence, manufacture and corrosion of iron
The Solvay process
Redox
Acids and bases
Nuclear chemistry
Petrochemicals

Associated titles in the Instant Lessons series

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